Gradual or Abrupt? Reform Pace, Advance Information, and Tax Compliance *

Andres Barinas-Forero† UCSD
Mariana Blanco‡ University of Turin
Andrea Lopez-Luzuriaga§ IDB
Carlos Scartascini¶ IDB

Abstract

How does the pace of a tax reform, and what taxpayers are told about it in advance, shape compliance? We report a laboratory experiment in which the statutory rate rises from 10% to 40% either in a single step or in three increments, crossed with whether participants learn the entire path at the outset or discover each change only when it takes effect. The two paths are constructed to carry the same average statutory rate, so any difference between them reflects behavior rather than the schedule itself. Compliance falls as the rate rises, but at the terminal 40% rate it is statistically indistinguishable across paths. Advance information matters instead: informed participants are 4 to 7 percentage points less likely to declare zero income and report about 5 percentage points more of their income. This effect is absent in the first block, when both groups face the same rate, and no change has yet occurred, which is inconsistent with intertemporal optimization and points to a tax-morale channel. Because gradualism yields no compliance dividend once the target rate is reached, the periods spent at intermediate rates are forgone revenue: governments should move to the target rate directly, and announce the move in advance.

Keywords: Tax compliance, Tax evasion, Tax morale, Fiscal reform, Laboratory experiment, Policy design.
JEL Classification: H21, H26, C91

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† University of California, San Diego, San Diego, CA, USA. Email: abarinas@ucsd.edu

‡ Department of Economics, University of Turin, Turin, Italy. Email: mariana.blanco@unito.it

§ Fiscal Management Division, Inter-American Development Bank, Washington, DC, USA. Email: andrealo@iadb.org

¶ Research Department, Inter-American Development Bank, Washington, DC, USA. Email: carlossc@iadb.org. ORCID: 0000-0001-7031-3983

1 Introduction

Raising tax revenue is a central policy problem across the developing world and in much of the advanced economy, yet tax hikes routinely stall in the legislature or meet swift public opposition, with even minor changes provoking resistance and occasionally civil unrest.[1] Because reforms are difficult to enact, governments are advised not only on what to legislate but on how: whether to move to the target rate in one step or several, and how much to tell taxpayers in advance. A substantial literature examines when reforms occur: their timing relative to the electoral calendar (Alesina et al., 2006; Hallerberg and Scartascini, 2017), the configuration of domestic political institutions (Ardanaz et al., 2020; Giuliano et al., 2013; Sturzenegger and Tommasi, 1998; Tommasi et al., 2014), and the role of crises in unlocking them (Alesina et al., 2006; Drazen and Easterly, 2001; Drazen and Grilli, 1993; Hallerberg and Scartascini, 2015). Much less is known about whether the design of the reform path itself changes how taxpayers behave once the reform is in force.

The choice between a single large reform and a sequence of smaller ones is a long-standing question in the political economy of reform, and the theoretical arguments cut in both directions. Gradualism can build support by revealing information about a reform's consequences as it unfolds and by creating constituencies that sustain the process, but it also leaves the reform exposed to reversal before completion (Dewatripont and Roland, 1992, 1995; Martinelli and Tommasi, 1997; Rodrik, 1996; Wei, 1997). One-time, extensive reforms concentrate attention and political costs; several minor reforms may attract less attention and prove easier to pass (Cottarelli, 2012). Information about the schedule of changes is similarly double-edged. Surprise may forestall organized collective action, but it can also exacerbate public distrust and derail current and future reforms (Keefer and Scartascini, 2022); advance information may help opponents organize, yet it can raise trust in and perceived legitimacy of the reform (Alessandro et al., 2021). These arguments all concern whether reforms are adopted. Our question is the one that follows adoption: conditional on a reform being implemented, does the path taken to the new rate, or advance knowledge of that path, change taxpayers' willingness to comply?

Whether the pace and the announcement of a reform matter is understudied because political and legal constraints prevent governments from implementing the same reform in different ways. Yet the question is consequential, because governments are routinely advised to follow one path rather than another. Japan, for example, was advised to raise the consumption tax from 5% to 15% through a sequence of increases, announced ahead of time: "Experiences elsewhere, as well as the specific circumstances of Japan, suggest that the strategy for raising the consumption tax be guided by the "four Ss"—it should start **S**ooner rather than later, be raised by **S**tepwise increases, **S**ustained for some time, and retain the very **S**imple current structure of the

[1] For instance, an attempt to increase the metro fare in Chile in 2019 triggered prolonged protests, resulting in significant infrastructural damage and major economic setbacks. Similarly, opposition to a tax hike in Colombia in 2021 led to numerous casualties.

consumption tax" (Kang et al., 2011, 3). This recommendation bundles two distinct instruments, a stepwise path and advance announcement, and treats both as ingredients of success. The subsequent experience did not bear the prediction out: the rate never moved beyond 10%, and reaching even that level required introducing substantial distortions into the tax. Whether the bundle disappointed because gradualism does not deliver, because announcement does not deliver, or because of the political circumstances in which the two were combined, cannot be settled from the Japanese case alone. Separating them requires variation that governments do not generate.

We generate that variation in the laboratory. The lab offers a decisive advantage here: it allows us to randomize a reform path that cannot be randomized in the field and can hardly be recovered from observational data, since countries that stagger their reforms differ systematically from those that do not. Building on the standard income-generation and tax-evasion paradigm (Alm and Malézieux, 2021; Friedland et al., 1978), we implement a 2×2 between-subjects design. Participants generate income through a real-effort task and then decide how much of it to declare, over forty periods grouped into four blocks. The statutory rate rises from 10% to 40% either in a single step (the *abrupt* design) or through intermediate rates of 20% and 30% (the *stepwise* design). Crossed with this, participants either learn the entire path of rates before the first period (the *information* condition) or encounter each change as a surprise when it takes effect (the *no information* condition). Two features of the design deserve emphasis. First, both paths carry the same average statutory rate over the experiment, 25%, and both start and end at the same rate; any difference in behavior between them therefore reflects the sequencing itself rather than a difference in the burden imposed. Second, because the environment offers no mechanism for protest, negotiation, or reversal, opposition to the reform can express itself only through reduced compliance and reduced effort. This is a narrowing of the real-world response set, but a deliberate one: it isolates the taxpayer-behavior channel from the political channels that ordinarily confound it.

To organize these predictions, we extend the canonical model of Allingham and Sandmo (1972) and Srinivasan (1973) to include costly income generation and an intrinsic motivation to pay taxes (Dwenger et al., 2016), and we embed it in a finite-horizon setting in which the rate follows a known or unknown path. Absent any dependence of intrinsic motivation on the history of rates, the model predicts identical behavior under gradual and abrupt paths once the terminal rate is reached: what matters is the rate in force, not the route to it. Advance information can nonetheless operate through two channels. The first is intertemporal. Although our payment protocol pays only two randomly selected rounds per block and thus removes any mechanism for shifting declared income across periods, subjects still maximize over total experimental earnings, so each round's gamble is background risk for every other. Informed participants anticipate both lower future earnings and larger future audit exposure, since the penalty is proportional to the rate; under decreasing absolute risk aversion, both considerations imply that informed participants should comply *more* in the first block, before any change has occurred. The second channel is motivational: transparency about the future path may raise trust in the

authority and hence tax morale (Feld and Frey, 2007; Keefer et al., 2021; Luttmer and Singhal, 2014), while unannounced shocks may erode it. The two channels are distinguishable because only the first predicts a difference in the first block. Online Appendix A.1 develops the model and states the hypotheses formally.

Consistent with the existing literature, higher rates depress reporting. Relative to the initial 10% rate, the 40% rate reduces the compliance rate by 8.8 percentage points, roughly twice the reduction observed at 20%. The pace of the increase, by contrast, does not matter: reaching 40% produces the same level of evasion whether the increase arrived in one step or three. Advance information does matter. Participants informed of the entire path are 4 to 7 percentage points less likely to declare no income at all at the 40% rate, depending on specification, and report about 5 percentage points more of their income. The extensive-margin result is robust to session-level clustering, individual-level clustering, a wild cluster bootstrap, and randomization inference; the intensive-margin result is robust to all but individual-level clustering.

The design lets us adjudicate between the two channels. In the first block, every arm faces the same 10% rate, and no change has yet occurred, so the intertemporal channel predicts higher compliance among the informed while the motivational channel predicts nothing. We find nothing: the four arms are statistically indistinguishable. The substantial effect we observe at the 40% rate cannot be intertemporal optimization. What remains is the motivational channel: participants who were told what was coming respond to the same rate increase with more compliance than participants who were not.

The paper makes three contributions. First, it extends the experimental literature on tax compliance (Alm, 2019; Alm and Malézieux, 2021) to a dimension it has not addressed. That literature has established how compliance responds to the level of the rate (Alm et al., 1992a), to audit probabilities and penalties, and to uncertainty about the enforcement environment (Alm et al., 1992b; Scotchmer and Slemrod, 1989), but it has treated the rate as a parameter rather than as something governments change along a path. We introduce reform paths and advance information about them, and show that the second matters while the first does not.

Second, it speaks to the political economy of reform sequencing. Theories of gradualism rest on the premise that a slower path eases acceptance (Dewatripont and Roland, 1992; Wei, 1997). In an environment where acceptance can be expressed only through compliance, we find no such easing. This does not refute those theories, which concern adoption rather than compliance, but it does locate a boundary: whatever gradualism buys, it does not buy taxpayer cooperation at the target rate.

Third, it contributes to work on information and public policy. Closest is the literature on anticipated versus surprise fiscal changes and their effects on economic activity (House and Shapiro, 2006; Mertens and Ravn, 2012; Ricco, 2015); adjacent is work on forward guidance and expectation formation in monetary policy (Jiang and Huang, 2023; Meade et al., 2015), where a surprise tightening does not have the same effect as an anticipated one (Jarociński and Karadi, 2020) and where the response to a surprise depends on the prevailing level of uncertainty

(De Pooter et al., 2018). Our results carry the analogous message for tax policy: how a reform is communicated matters at least as much as how it is phased. We note that our own design does not hold anticipability constant—the move from 30% to 40% was arguably easier to foresee than the move from 10% to 40%—and randomizing the unexpectedness of a change while holding its magnitude fixed remains open for future work.

Because our evidence comes from a student laboratory sample, its external validity warrants comment. Laboratory tax experiments reproduce the qualitative comparative statics found in field settings, and the compliance behavior of students and of actual taxpayers responds similarly to the policy parameters we manipulate (Alm et al., 2015; Choo et al., 2016; Mascagni, 2018). Our own distribution of compliance, bimodal with substantial mass at both full compliance and full evasion, matches both the laboratory meta-analysis of Alm and Malézieux (2021) and administrative data (Carrillo et al., 2021; Castro and Scartascini, 2015). We read our estimates as evidence about the direction and relative magnitude of the two treatments rather than as calibrated predictions of field responses.

Finally, the results carry a policy implication that is worth stating carefully, because it does not follow from a revenue comparison within our experiment. Our two paths were constructed to carry the same average statutory rate, so which of them collects more in the laboratory is an artifact of the schedules we chose rather than a finding. What the experiment establishes is that compliance at the terminal rate does not depend on the route taken to reach it. That invariance is what the policy question turns on. If arriving at 40% by way of 20% and 30% yields no compliance dividend once 40% is in force, then the periods spent at intermediate rates are revenue forgone for nothing, and a government that has determined it needs the higher rate should legislate the higher rate rather than phase it in. We observe path-invariance of compliance between a three-step path and a one-step path taken midway through the experiment, and extrapolate that it would also hold for a path taken sooner. Advance announcement, by contrast, is not a matter of indifference: informing taxpayers of what is coming measurably reduces the compliance loss that accompanies the increase. Returning to the advice given to Japan, our evidence supports announcing the change and moving sooner, but not the recommendation to move stepwise.

2 Experimental Design

Testing whether the pace of a reform and advance information about it affect compliance requires holding the reform's destination fixed while varying the route and what taxpayers know about it. No government generates that variation, so we construct it in the laboratory. The experiment was programmed in oTree (Chen et al., 2016) and conducted with 410 college students from several universities in Bogotá, Colombia, recruited through the Rosario Experimental and Behavioral

Economics Lab (REβEL) between November 4, 2022, and March 2, 2023.[2] The instructions, presented in Spanish, are reproduced in the Online Appendix (Figures A.3–A.18).

2.1 Real-Effort Task (RET)

In the first part of the experiment, participants engaged in an encryption task as described by Erkal et al. (2011), which lasted for 5 minutes. Each participant was given a unique encryption table that assigned a random numerical value to each letter of the alphabet, with the letters arranged in a randomized sequence to ensure non-sequential and unpredictable assignments. The main task was to encrypt a 3-letter word using the provided table, substituting each letter with its corresponding number. After each word was correctly encoded, the encryption table changed, and a new word was presented. Each correctly encrypted word earned the participant one experimental point (EP).[3] Once the 5 minutes ended, participants learned about their performance and earnings. Overall, this task allowed participants to generate a non-taxed income and also allowed us to measure their baseline ability for this encryption task.

2.2 Tax Evasion Game (TEG)

In the second part of the experiment, participants engage in a 40-period tax evasion game (Alm and Malézieux, 2021) divided into four (4) blocks of 10 periods each. Each block begins with a 5-minute real-effort task (RET) that allows participants to generate income. The structure of each real-effort task implemented in the TEG follows the one presented in Section 2.1 with the same word length and conversion fee per word correct. If that block is subject to a tax rate change, this is informed right before the start of the RET; they are also reminded that afterward they will have to decide how much of their income they would like to declare. Once they have learned their generated income z_t, the tax evasion game (TEG) begins and is played for 10 independent rounds.

In each round of the TEG, participants report their income (the tax rate is also visible on this screen), and the tax paid is calculated based on the reported income y_t. It is worth noting that individuals are permitted to underreport their income, as the instructions do not stipulate that reporting must be accurate. Moreover, participants are informed that there is auditing and a penalty that all audited individuals must pay if they are caught under-reporting, which suggests that under-reporting is an option.

After reporting their income, each participant pressed a button to draw a random number determining whether they would be audited in that round. Participants knew, including on

[2] The experimental protocol was approved by the Institutional Review Board (IRB) at Universidad del Rosario, CS-422.

[3] Each experimental point was valued at a fee of $0.05 USD.

the audit screen itself, that the audit probability was 20% and constant across periods and treatments.[4] If audited, the computer compared total and reported income, and any shortfall triggered a penalty equal to four times the evaded tax: $\text{Penalty}_t = 4 \times \tau_t(\text{Total income}_t - \text{Reported income}_t)$, where τ_t is the tax rate in period t and the penalty multiplier of 4 is constant across rounds and treatments. Both the tax collected and any penalty were deducted from the participant's period payoff.

Taxes collected from each participant were donated to a non-political, non-governmental organization that the participant selected from a fixed list at the end of the session. Participants were informed of this arrangement in the declaration instructions, before the first block. Directing revenue to a recipient the participant chooses is a departure from experiments in which collected taxes simply leave the participant's account, and it plausibly raises the level of compliance we observe relative to a pure loss frame. Because the arrangement is identical across all four arms and constant across all forty periods, it cannot generate treatment effects, but it does mean that the levels we report should be read as compliance in an environment with a visible and non-governmental use of funds.

2.3 Treatment arms

We implement a 2×2 between-subjects design to identify the effect of the pace of the reform and of advance information about that pace. Two tax-rate trajectories take participants from an initial rate of 10% to a terminal rate of 40%: one through intermediate rates of 20% and 30% (the **stepwise** design) and one through a single jump (the **abrupt** design). Table 1 gives the trajectories by block. The two paths are constructed to be comparable in three respects that matter for interpretation. They begin at the same rate, they end at the same rate, and they impose the same average statutory rate over the forty periods, 25% in both cases. The last of these is what allows a clean reading of the pace comparison: any difference in behavior between the designs reflects the sequencing of the increase rather than a difference in the total burden imposed. It also means that in-sample revenue differences between the two designs are a mechanical consequence of the schedules we chose rather than a behavioral result, a point we return to when drawing policy implications.

The information dimension has two levels. In the **information** treatment, participants were shown the complete trajectory of tax rates before the first block, so that the pace and depth of the reform were known from period 1. In the **no information** treatment, participants learned each new rate only when it took effect, so each increase arrived as a surprise. The announcement took the form of a single screen presented after the general instructions and before the first real-effort

[4] Holding the audit probability constant isolates the tax-rate manipulation from the effects of institutional uncertainty about enforcement (Alm et al., 1992b; Kamm et al., 2021). Before the first block, participants answered two incentivized-format control questions requiring them to compute the payoff of a hypothetical participant under audit and no audit; participants could not proceed until both were answered correctly.

Table 1 - Designs on the **pace** of the reform

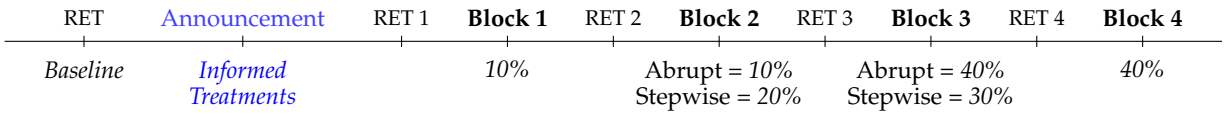
	Tax rate (%)				
	Block 1	**Block 2**	**Block 3**	**Block 4**	Average
Abrupt design	10%	10 %	40%	40%	25%
Stepwise design	10%	20 %	30%	40%	25%

task; no other aspect of the protocol differed across information conditions.

This manipulation bundles several things that a real announcement also bundles, and it is worth being explicit about what they are. Informed participants know the sequence of rates; they also know with certainty that rates will change, which uninformed participants do not; and they receive a signal that the authority is willing to commit to and disclose its plans in advance. Our design cannot separate these components, and we do not claim to identify the effect of the tax-path information alone. What it does identify is the effect of the bundle, which is the object policymakers actually choose over.[5]

The sequence of events displayed during the TEG is depicted in Figure 1 and the evolution of the tax rate in Figure 2.

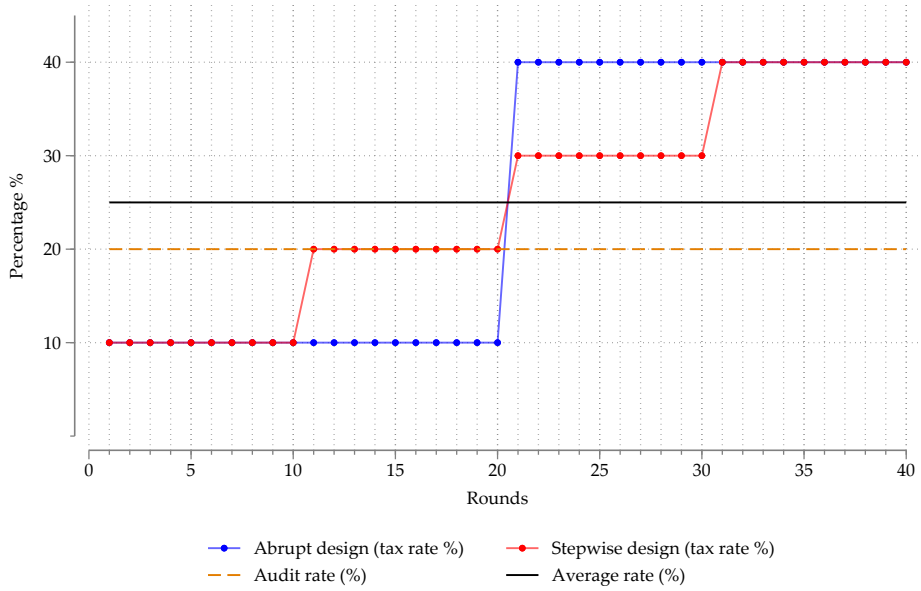
Figure 1 - Timeline of the experiment



Note: Experimental timeline from left to right. (RET) = Real Effort Task (Erkal et al., 2011). The announcement reflects our information treatment, where treated participants receive information on the tax reform design. Finally, before each real effort task, participants were informed about the new tax rate for the respective block. Under each tick is depicted the income tax rate for each treatment.

[5] One further consideration is that informed participants receive an additional screen from the experimenter emphasizing the tax schedule, which could in principle generate a demand effect. We report in Section 3 that the information effect is absent in the first block and appears only once rates have risen, a pattern that is difficult to reconcile with a general disposition to please the experimenter, and that it is not driven by the opening or closing rounds of a block.

Figure 2 - Tax reform design



Note: In blue, the income tax rate for the abrupt treatments is depicted. In red, the income tax rate for the stepwise treatments is illustrated. In black, the average income tax rate across the entire experiment, regardless of the treatment status (Average tax rate = 25%), is depicted. In orange (dashed), the audit rate (Audit rate = 20%) is depicted.

The 2x2 design results in 4 treatments based on the **pace** of the tax rate change and the **information** they received about that pace. Table 2 presents the resulting treatments. Out of our total of 410 participants, 205 received information about the trajectory of the tax rate, while the other 205 were not informed about the structure of the reform. Additionally, 206 subjects participated in the abrupt design, while 204 participated in the stepwise design.

Treatment status was assigned at the session level, so all participants within the same session received the same treatment.[6] On average, each of the 16 sessions had 25 participants. Finally, at the end of the TEG activity, participants completed a socio-demographic questionnaire, decided the NGO they wanted their collected taxes to go to, and provided the necessary information for payment.

[6] Throughout the paper, standard errors are clustered at the session level rather than at the individual level, since treatment was assigned at the session level (16 sessions, 4 per treatment arm) and all participants within a session share the same randomly assigned treatment. This follows the standard guidance to cluster at the level of treatment assignment (Abadie et al., 2017). Because our design has only 16 clusters in total (as few as 4 per arm in some comparisons) session-clustered asymptotic standard errors alone may not be reliable, so we complement them with a wild cluster bootstrap (Cameron et al., 2008) using Webb's six-point weights, which are better suited than the standard Rademacher weights when the number of clusters on one side of a comparison is this small (MacKinnon and Webb, 2018), wherever we report information-treatment effects in the main text. Tables A.4 and A.5 from the Appendix repeat our two headline compliance-outcome specifications clustering standard errors at the individual level instead. We complement these with randomization inference that resamples entire sessions to treatment status, reported alongside our main estimates.

Table 2 - Treatments 2 × 2 between subjects design

	Information	
Pace	No information	Information
Abrupt	No information - Abrupt (C) (n_c =**101**)	Information - Abrupt (T_1) (n_1 =**105**)
Stepwise	No information - Stepwise (T_2) (n_2 =**104**)	Information - Stepwise (T_3) (n_3 =**100**)

Note: Total sample = 410 subjects. In parentheses, the abbreviation of the treatment. In (**bold**), the number of participants assigned to the treatment.

Table A.1 reports summary statistics and tests of balance across the four arms. Participants averaged 21.5 years of age, 46.5% were female, and 13% were graduate students; the distribution of socioeconomic stratum, weekly expenses, risk aversion, and political preferences is reported in the table. Of the 24 characteristics-by-treatment comparisons, only one differs significantly at the 5% level and 2 at 10%. We address all potential differences by adding controls in all specifications for baseline performance and demographic characteristics. Our conclusions are unchanged.

2.4 Payment

Participants' earnings comprised three components: a participation fee of approximately $2.50 USD, the untaxed income generated in the initial *Baseline* encryption task, and the payoff from the tax evasion game. Participants encrypted an average of 22 words in the baseline task, worth approximately $1.10 USD. For the tax evasion game, participants knew in advance that only two randomly selected rounds per block would be paid, a design choice intended to keep incentives independent across rounds and blocks and to prevent wealth accumulated in early rounds from affecting later decisions. At the end of the activity, the computer drew two periods per block, and the payoffs realized in those eight periods constituted the game payoff. Participants earned $12 USD on average, roughly 8.4 times the Colombian hourly minimum wage of $1.42 USD for a session of approximately 90 minutes. All payments were made at the end of the session.

Following the tax evasion game, participants completed a socio-demographic questionnaire, selected the organization to receive their collected taxes, and provided payment details. The questionnaire elicited risk aversion by asking how willing the participant was to take risks (0 = not at all, 10 = very willing), along with age, gender, socioeconomic stratum, weekly expenses, and political preferences. These variables constitute the control vector used throughout Section 3.

3 Results

3.1 Empirical Strategy

We begin by describing how compliance covaries with the level of the tax rate, pooling across treatment arms:

$$y_{i,t} = \alpha + \sum_{k \in \{20,30,40\}} \beta_k \mathbb{1}[\tau_t = k] + X_i'\gamma + \varepsilon_{i,t} \tag{1}$$

where $y_{i,t}$ is the compliance rate (the share of generated income participant i declares in round t), τ_t is the tax rate in round t with the 10% rate as the omitted category, and X_i is a fixed vector of individual controls: age, gender, socioeconomic stratum, weekly expenses, risk aversion, political preferences, and baseline encryption performance. The same vector is used in every specification reported in the paper; specifications adding an indicator for having been audited in the previous round and lagged compliance are reported in the Online Appendix. We estimate Equation (1) on the full sample and separately within each pace-by-information arm. Standard errors are clustered at the session level, the unit at which treatment was randomly assigned; the Online Appendix discusses this choice and its robustness to clustering at the individual level and to a wild cluster bootstrap.

Since our design varies two treatment dimensions: the **pace** of the reform (abrupt vs. stepwise) and whether participants received advance **information** about the entire tax rate path, we next ask how these two dimensions jointly shape the tax-rate effect just described, focusing on the two tax-rate blocks common to both designs: block $m = 1$ (10% tax rate) and block $m = 4$ (40% tax rate). We estimate the effect of both dimensions jointly using three treatment-arm dummies. We keep the abrupt, uninformed arm as the reference category:

$$\begin{aligned} y_{i,t} = \alpha &+ \beta_1 \,\text{AbruptInfo}_i + \beta_2 \,\text{StepwiseNoInfo}_i + \beta_3 \,\text{StepwiseInfo}_i \\ &+ X_i'\gamma + \delta_t + \varepsilon_{i,t}, \qquad t \in \text{block } m \end{aligned} \tag{2}$$

where $y_{i,t}$ is now either the compliance rate or an indicator for full evasion (declaring zero income) in round t, and δ_t denotes round fixed effects, included in a subset of specifications.[7]

[7] Inference deserves comment before we present results. Treatment was assigned at the session level, so we cluster standard errors on sessions throughout (Abadie et al., 2017). With sixteen sessions in total, and as few as four per arm, asymptotic cluster-robust standard errors are unreliable on their own. We therefore report three additional procedures alongside every main estimate. The first is a wild cluster bootstrap with Webb six-point weights and 9,999 replications (Cameron et al., 2008; MacKinnon and Webb, 2018), which is better suited than Rademacher weights when the number of clusters is this small. The second is randomization inference, which resamples entire sessions into treatment status exactly as the original randomization did and compares the observed estimate to

Because our sample size affords limited power to detect effects within each of the four arms separately, as discussed below, we also pool across the pace dimension and estimate the effect of information alone, combining the abrupt and stepwise arms within each information status:

$$y_{i,t} = \alpha + \beta\,\text{Information}_i + X_i'\gamma + \delta_t + \varepsilon_{i,t}, \qquad t \in \text{block } m \tag{3}$$

where Information_i is an indicator equal to one if participant i was informed in advance of the entire tax rate path.

We focus on two dimensions of taxpayer behavior: the effort to generate income that could be reported, and the actual compliance decision. We begin by examining the effect of the tax rate increase and our treatments on effort. We then turn to compliance. First, we examine the overall effect of the tax rate increase on compliance and analyze the effects of the **information** and **pace** of the reform by comparing results across treatment arms to understand how the compliance decline associated with higher tax rates can be mitigated through the design of the reform.

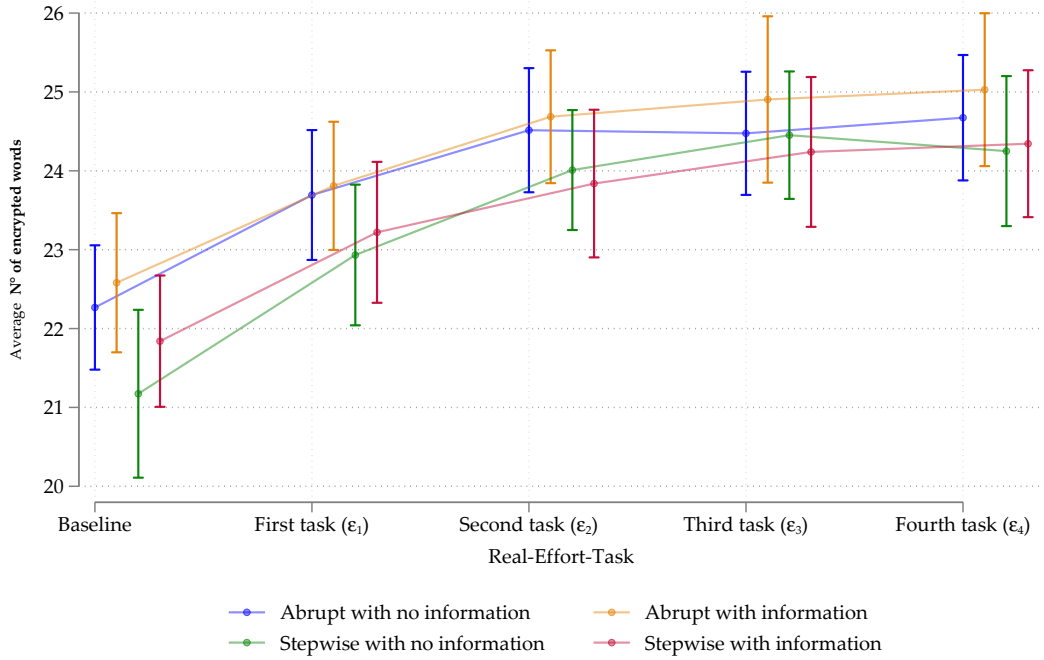
3.2 Effects on Effort

We begin with effort. Table A.3 compares performance across treatment arms within each real-effort task, holding constant the position of the task in the session; this is the comparison that isolates the treatments, since arms are randomly assigned. We find no systematic differences: neither the pace of the reform nor advance information about it affects the number of words participants encrypt in any block. Table A.2 reports the corresponding across-task comparison, but we caution against reading its coefficients as tax-rate effects. As Figure 3 makes clear, performance rises steeply from the baseline task to the first task and then flattens in every arm, informed and uninformed, abrupt and stepwise alike. The across-task coefficients in Table A.2 are therefore predominantly a learning curve, and they are of similar magnitude in arms facing very different rate paths.

The absence of an effort response is expected given our setting rather than informative about labor supply. Participants could evade, and the audit probability was fixed at 20%, so the cheaper margin of adjustment to a higher rate was reporting rather than production. More fundamentally, time in the laboratory carries no explicit opportunity cost: participants had no alternative activity, so exerting effort weakly dominated regardless of the rate. Our design is not informative about the labor-supply response to taxation, and we do not interpret it as such; the compliance results that follow are where the treatments operate.

the resulting distribution over 1,000 draws; this procedure is exact under the sharp null and makes no asymptotic appeal, and we regard it as the most credible of the three. The third is clustering at the individual level, reported in the Online Appendix, which we view as anti-conservative given the assignment mechanism but report because it bounds our results in the other direction. Where these procedures disagree, we say so.

Figure 3 - Performance in RET by tasks (ϵ)

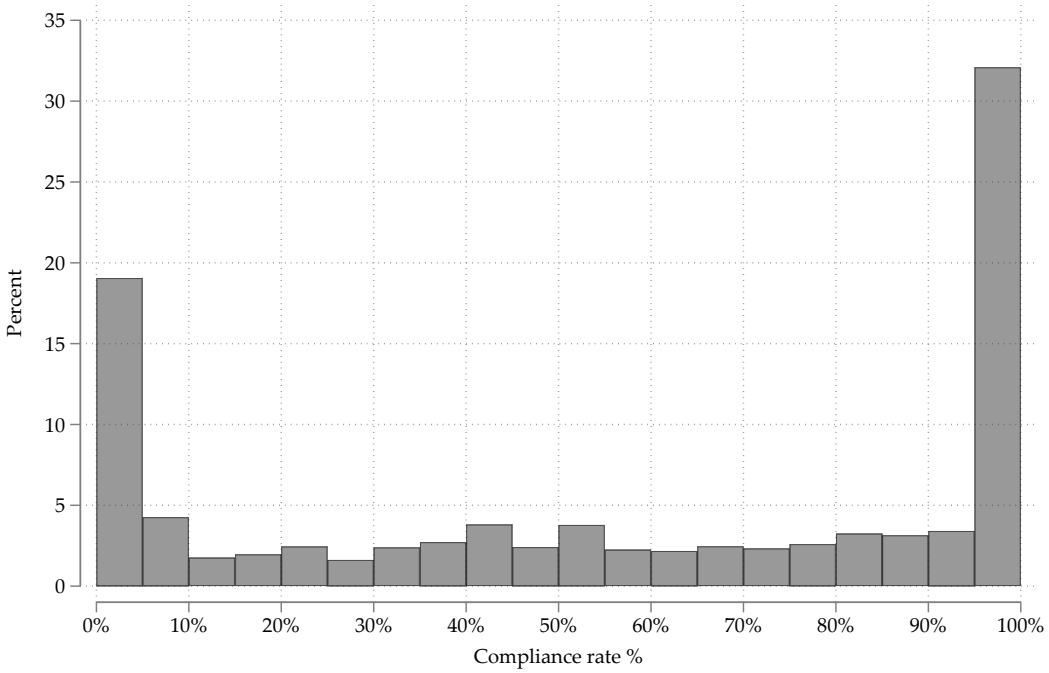


Notes: **Baseline** denotes the first real effort task. The first task (ϵ_1) corresponds to the real effort task under a tax rate of 10%. The second task (ϵ_2) corresponds to: for *Stepwise* treatments, a tax rate of 20%; and for *Abrupt* treatments, a continued rate of 10%. The third task (ϵ_3) corresponds to: for *Stepwise* treatments, a tax rate of 30%; and for *Abrupt* treatments, a jump to 40%. The fourth task (ϵ_4) corresponds to: for *Stepwise* treatments, a tax rate of 40%; and for *Abrupt* treatments, a second task at 40%. 95% confidence intervals are shown.

3.3 Effect on Compliance

We now turn to compliance. Pooling across periods and treatments, Figure 4 presents the distribution, which is strongly bimodal: full compliance and full evasion coexist under identical experimental parameters, a pattern found both in laboratory meta-analysis (Alm and Malézieux, 2021) and in administrative data (Carrillo et al., 2021; Castro and Scartascini, 2015). Participants declared their entire income in 33% of decisions and declared nothing at all in 18% of decisions; the average compliance rate is 57% (SD = 0.39). No single-parameter expected-utility account rationalizes both modes at the same audit probability and penalty, which is the standard motivation for admitting behavioral responses (loss aversion, overconfidence, and tax morale) alongside the deterrence parameters (Luttmer and Singhal, 2014; Slemrod, 2019). The bimodality is also the reason we analyze the extensive and intensive margins separately below.

Figure 4 - (*Full sample*) Histogram of compliance



Note: Compliance rate (%) denotes the fraction of endowment declared by the participant. Compliance rate of 0% denotes **total evasion**, while 100% denotes **full compliance**. The histogram contains all decisions made by participants across all rounds. Full compliance of our sample = 33% (Full compliance in other settings = 45% (Alm and Malézieux, 2021)). Full evasion of our sample = 18% (Full evasion in other settings = 19% (Alm and Malézieux, 2021))

Table 3 presents the average compliance rate across all periods by treatment. The first column shows results for the full sample, while the remaining columns display results by treatment arm. Across specifications, higher tax rates are associated with lower compliance rates.

When the sample is pooled across treatments, the mean compliance rate under the 10% tax rate is 61.8%. Relative to this baseline, increasing the tax rate to 20% and 30% leads to statistically significant reductions in compliance of 4.3 and 4.5 percentage points, respectively, although the wild cluster bootstrap p-values for these two intermediate steps (0.069 and 0.048, respectively) indicate that this evidence is weaker than the asymptotic significance stars alone suggest, given our small number of independent sessions. When the tax rate increases to 40%, the decline becomes more pronounced, with a reduction of 8.8 percentage points, a result that is highly robust under the wild cluster bootstrap ($p<0.001$).

The pattern across arms is informative. The two arms that received advance information show the smallest declines in compliance at the 40% rate: 7.0 percentage points in the stepwise design and 7.2 in the abrupt design. The two uninformed arms show the largest: 9.0 percentage points in the abrupt design and 13.5 in the stepwise design. The arms sort by information rather than by pace: the two informed arms are nearly identical to each other, whereas neither pair of same-pace arms is.

Given the bimodality of the compliance distribution, we examine two outcomes that capture

Table 3 - Effect of tax rate over **compliance** rate

Dep Var: **Compliance rate**	General sample	Stepwise		Abrupt	
		No information	Information	No information	Information
	(1)	(2)	(3)	(4)	(5)
20% tax rate	-0.043**	-0.071**	-0.062*		
	(0.020)	(0.021)	(0.026)		
30% tax rate	-0.045**	-0.092***	-0.043		
	(0.018)	(0.014)	(0.026)		
40% tax rate	-0.088***	-0.135***	-0.070*	-0.090***	-0.072**
	(0.009)	(0.010)	(0.027)	(0.011)	(0.014)
Mean compliance with 10% tax rate	0.618	0.649	0.630	0.581	0.632
Controls	✓	✓	✓	✓	✓
Clusters	16	4	4	4	4
Observations	15901	4037	3850	3939	4075
Adjusted R-squared	0.04	0.06	0.09	0.03	0.05
p-value H_0 : 20% tax rate = 30% tax rate	0.840	0.100	0.070		
p-value H_0 : 20% tax rate = 40% tax rate	0.036	0.077	0.783		
p-value H_0 : 30% tax rate = 40% tax rate	0.013	0.115	0.248		
WCB p (20% tax rate)	0.069	0.082	0.170		
WCB p (30% tax rate)	0.048	0.057	0.282		
WCB p (40% tax rate)	0.000	0.025	0.165	0.032	0.032

Notes: *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. We include controls for female, age, stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization) are presented in parentheses; Column (1) pools all 16 sessions, Columns (2)–(5) use the 4 sessions in the relevant arm. The *WCB p-value* rows report wild cluster bootstrap p-values (Webb weights, 9,999 replications) for the 20%, 30%, and 40% tax rate coefficients, which account for the small number of clusters.

distinct decisions. The first is an indicator for full evasion, equal to one when a participant declares zero income; this captures the extensive-margin decision of whether to participate in the reporting system at all. The second is the compliance rate itself, the fraction of generated income declared, which captures the overall degree of reporting. We note that the compliance rate is unconditional and therefore mixes both margins.

Providing information about the reform trajectory reduces the probability of full evasion, with effects that are most pronounced when tax rates are high. Table 4 and Figure 5 present the effects of our treatment arms using the abrupt with no information treatment as the baseline for a comparison on the likelihood of full evasion, disaggregated by tax rate. Columns (1)–(4) report estimates for a 10% tax rate, while Columns (5)–(8) show results for the 40% tax rate. The outcome variable is a binary indicator equal to one if participants declared zero income.

At the 10% rate, treatment effects are small and statistically insignificant in every specification, with mean full evasion around 10%. At the 40% rate, full evasion doubles to 20%, and the informed arms diverge from the uninformed ones. Participants who received advance information are less likely to declare zero income, by 7.8 percentage points in the abrupt design and 7.2 in the stepwise design without controls, and by 3.5 and 3.7 percentage points respectively once controls and round fixed effects are added. The effect is of similar magnitude under both reform paces, and tests of equality between the two informed arms cannot reject at any conventional level ($p = 0.85$ to 0.99 across specifications), while tests comparing the informed abrupt arm to the

uninformed stepwise arm reject at the 1% level. Transparency about upcoming changes helps sustain participation in the reporting system at the extensive margin, and it does so regardless of how the increase is phased.

The first block provides a test that separates the two channels through which information might operate. In block 1 every arm faces the same 10% rate, no change has yet occurred, and the only difference between informed and uninformed participants is knowledge of what is coming. The intertemporal channel makes a signed prediction here (H3 in Online Appendix A.1): informed participants anticipate both lower future earnings and larger future audit exposure, and under decreasing absolute risk aversion each implies that they should comply *more* in block 1 than uninformed participants. The motivational channel predicts nothing in block 1, because no shock has yet been delivered and no transparency has yet been vindicated or betrayed.

We find no anticipation effect. The pooled information coefficient in block 1 is 0.005 (s.e. 0.026) for the compliance rate (Table 7) and -0.028 (s.e. 0.02) for full evasion (Table 6), neither approaching significance under any inference procedure; randomization inference returns $p = 0.85$ for the compliance rate, with 847 of 1,000 resampled draws exceeding the observed estimate (Figure A.2a). All four arms are statistically indistinguishable from one another. The null is unsurprising on calibration grounds: at total stakes near \$12, wealth effects across rounds are negligible for any plausible utility function, and the round-by-round audit lottery encourages participants to evaluate each decision in isolation. But it is informative, because the same pooled comparison detects a substantial effect at the 40% rate. Whatever advance information does, it does not operate by inducing participants to reallocate compliance across periods.

Figure 5 - Probability of full evasion across rounds by treatment



Table 4 - Effect of each treatment over **full evasion**

Dep Var: **Full evasion**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Abrupt with information (T_1)	-0.051	-0.036	-0.036	-0.041	-0.078**	-0.035*	-0.035*	-0.063**
	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)
Stepwise with no information (T_2)	-0.026	-0.018	-0.018	-0.019	-0.003	0.009	0.010	0.000
	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Stepwise with information (T_3)	-0.031	-0.015	-0.015	-0.030	-0.072*	-0.037	-0.037	-0.063*
	(0.04)	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Mean Dep. Variable	0.10	0.11	0.11	0.11	0.20	0.20	0.20	0.20
Effect size (%)	-48.3	-33.4	-33.4	-37.3	-39.8	-18.0	-18.0	-32.3
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4100	3681	3681	3690	4060	4041	4041	4060
Clusters	16	16	16	16	16	16	16	16
Adjusted R-squared	0.00	0.13	0.13		0.01	0.17	0.17	
p-value $H_0 : T_1 = T_2$	0.280	0.303	0.304	0.333	0.005	0.019	0.019	0.007
p-value $H_0 : T_1 = T_3$	0.403	0.264	0.265	0.692	0.848	0.928	0.928	0.998
p-value $H_0 : T_2 = T_3$	0.816	0.878	0.877	0.673	0.040	0.116	0.117	0.050
WCB p (Abrupt Info)	0.331	0.386	0.386		0.059	0.129	0.129	
WCB p (Step No Info)	0.644	0.662	0.661		0.929	0.748	0.747	
WCB p (Step Info)	0.597	0.722	0.721		0.081	0.260	0.261	

Notes: *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. We include controls for female, age, stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses. The *WCB p-value* rows report wild cluster bootstrap p-values (Webb weights, 9,999 replications) for each treatment coefficient, which account for the small number of clusters; these are not reported for the LASSO column, since the wild cluster bootstrap does not support the double-selection LASSO estimator.

Note: The vertical axis is the share of decisions in which participants declared zero income (**full evasion**). In squares (■), **abrupt** treatments; in circles (●), **stepwise** treatments. Dashed lines, **informed** treatments; solid lines, **uninformed** treatments. **Left panel:** 10% tax rate. **Right panel:** 40% tax rate.

We repeat the same analysis for the compliance rate, which captures behavior at the intensive margin of tax compliance. At the 40% tax rate, providing information significantly increases the compliance rate in both the abrupt and stepwise designs (Table 5); neither effect is significant under individual-level clustering. Under the wild cluster bootstrap, the abrupt-design effect is only marginally significant ($p \approx 0.08$-0.09), while the stepwise-design effect is significant without controls ($p = 0.045$) but not once controls are added ($p \approx 0.13$). As with the pooled results, the pace of the reform itself does not generate significant differences in the compliance rate. See Figure A.1 for the corresponding trends.

An ex-post power calculation indicates that our sample is not large enough to detect effects of plausible size within each arm separately, particularly for the compliance rate, whose variance is substantially higher than that of the binary full-evasion measure.[8] Because participants were

[8] We compute the ex-post minimum detectable effect with `clustersampsi`, treating the session as the clustering unit since treatment was randomized at that level (16 sessions, 4 per arm). We collapse to one observation per

Table 5 - Effect of each treatment over **compliance** rate

Dep Var: **Compliance rate**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Abrupt with information (T_1)	0.029	0.008	0.008	0.032	0.076**	0.033**	0.033**	0.065**
	(0.04)	(0.02)	(0.02)	(0.04)	(0.03)	(0.01)	(0.01)	(0.03)
Stepwise with no information (T_2)	0.040*	0.018	0.018	0.043*	0.032	0.009	0.009	0.027
	(0.02)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Stepwise with information (T_3)	0.021	0.005	0.005	0.028	0.065**	0.027*	0.026*	0.066***
	(0.03)	(0.01)	(0.01)	(0.03)	(0.02)	(0.01)	(0.01)	(0.02)
Mean Dep. Variable	0.61	0.60	0.60	0.60	0.49	0.49	0.49	0.49
Effect size (%)	4.8	1.4	1.4	5.3	15.6	6.7	6.7	13.4
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4090	3681	3681	3681	4040	4039	4039	4040
Clusters	16	16	16	16	16	16	16	16
Adjusted R-squared	0.00	0.32	0.32		0.01	0.28	0.28	
p-value $H_0 : T_1 = T_2$	0.786	0.608	0.608	0.811	0.111	0.068	0.067	0.131
p-value $H_0 : T_1 = T_3$	0.831	0.830	0.830	0.933	0.696	0.517	0.512	0.964
p-value $H_0 : T_2 = T_3$	0.499	0.396	0.396	0.653	0.102	0.188	0.188	0.056
WCB p (Abrupt Info)	0.552	0.694	0.694		0.080	0.088	0.088	
WCB p (Step No Info)	0.189	0.310	0.309		0.245	0.647	0.649	
WCB p (Step Info)	0.496	0.732	0.731		0.045	0.133	0.134	

Notes: *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. We include controls for female, age, stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses. The *WCB p-value* rows report wild cluster bootstrap p-values (Webb weights, 9,999 replications) for each treatment coefficient, which account for the small number of clusters.

randomly assigned and the information effect is of similar magnitude under both paces, we pool across the pace dimension to estimate the effect of information alone.[9]

The pooled estimates show that advance information significantly mitigates the compliance loss associated with the higher rate, on both margins. At the extensive margin, information reduces the probability of declaring zero income by 4.1 to 7.4 percentage points depending on specification, a reduction of roughly a fifth to a third relative to the uninformed mean of 20% (Table 6). At the intensive margin, it raises the average compliance rate by approximately

participant (mean compliance in block 4) and estimate the intracluster correlation across sessions via a random-intercept model; the session-level ICC is indistinguishable from zero, though with sixteen sessions this estimate is itself imprecise. For a single-arm comparison (4 sessions, average cluster size $m = 25$) the minimum detectable effect with 80% power at $\alpha = 0.05$ is 0.14, against 0.12 under the counterfactual assumption that the individual is the relevant cluster. `clustersampsi` flags designs with fewer than ten clusters per arm as underpowered, which applies here.

[9] Pooling doubles the number of clusters to eight sessions per information status (average cluster size $m = 26$), reducing the minimum detectable effect at 80% power to 0.09. This is essentially the same as the 0.09 obtained under individual-level clustering, because the gain from doubling the number of clusters offsets the loss from clustering at the coarser level.

5 percentage points, or about 10% relative to the uninformed mean of 0.50 (Table 7). The extensive-margin result holds under session-level clustering, individual-level clustering, the wild cluster bootstrap ($p = 0.003$ to 0.020), and randomization inference. The intensive-margin result holds under session-level clustering, the wild cluster bootstrap ($p = 0.015$ to 0.041), and randomization inference ($p = 0.02$, with 21 of 1,000 resampled draws exceeding the observed estimate), but not under individual-level clustering, where it is significant at the 10% level only.

Table 6 Effect of the information treatment over **full evasion**

Dep Var: **Full evasion**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information treatment	-0.028	-0.017	-0.017	-0.023	-0.074***	-0.041**	-0.041**	-0.063***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Mean dep. var	0.09	0.10	0.10	0.09	0.19	0.20	0.20	0.19
Effect size (%)	-30.3	-17.7	-17.7	-24.3	-37.9	-21.0	-21.0	-32.5
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4100	3681	3681	3690	4060	4041	4041	4060
Clusters	16	16	16	16	16	16	16	16
Adjusted R-squared	0.00	0.13	0.13		0.01	0.17	0.17	
WCB p-value (Webb)	0.315	0.475	0.475		0.003	0.020	0.020	

Notes: Treatment arms are pooled across the pace (abrupt/stepwise) dimension, comparing participants who received advance information about the reform trajectory to those who did not. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses. The *WCB p-value* row reports the wild cluster bootstrap p-value (Webb weights, 9,999 replications), which accounts for the small number of clusters; these are not reported for the LASSO column, since the wild cluster bootstrap does not support the double-selection LASSO estimator. As shown in Appendix Table A.5, this result also holds under individual-level clustering.

Table 7 Effect of the information treatment over **compliance** rate

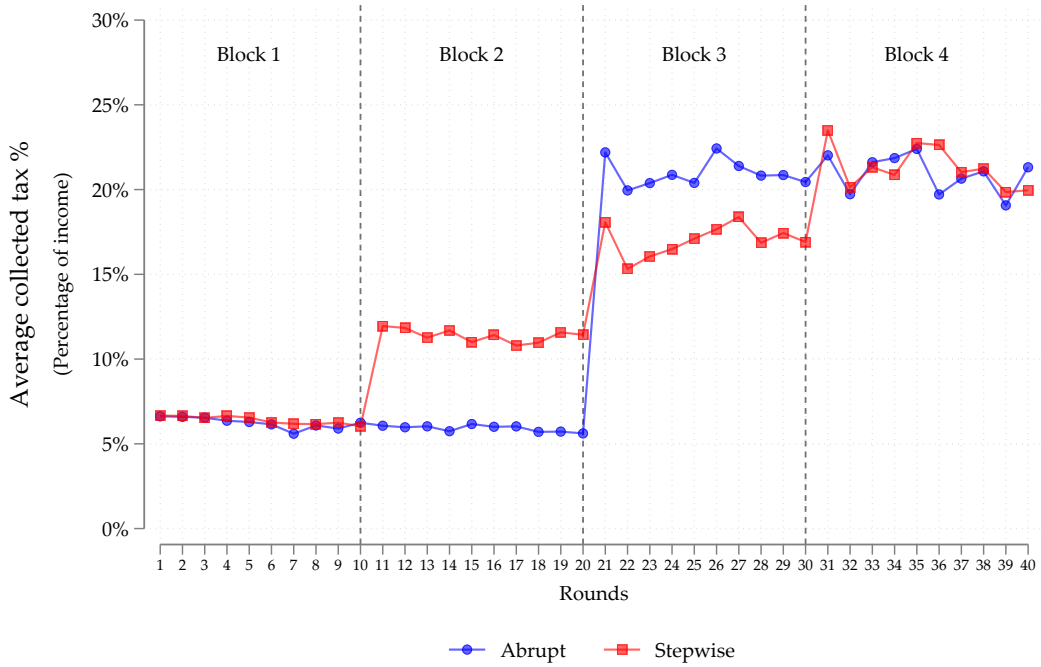
Dep Var: **Compliance rate**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information treatment	0.005	-0.002	-0.002	0.007	0.054**	0.048**	0.048**	0.054**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Mean dep. var	0.63	0.63	0.63	0.63	0.50	0.50	0.50	0.50
Effect size (%)	0.8	-0.4	-0.3	1.0	10.9	9.5	9.5	10.7
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4090	3681	3681	3681	4040	4040	4040	4040
Clusters	16	16	16	16	16	16	16	16
Adjusted R-squared	0.00	0.01	0.01		0.00	0.01	0.02	
WCB p-value (Webb)	0.852	0.938	0.938		0.015	0.041	0.041	

Notes: *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. We include controls for female, age, stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses. The *WCB p-value* row reports the wild cluster bootstrap p-value (Webb weights, 9,999 replications), which accounts for the small number of clusters; these are not reported for the LASSO column, since the wild cluster bootstrap does not support the double-selection LASSO estimator. Columns (6)–(7) (block 4, with controls) are significant at the session level and under the wild cluster bootstrap, but not under individual-level clustering; see the discussion in the Online Appendix.

Taken together, these findings indicate that the pace of implementation does not measurably affect compliance at the terminal rate, while advance information does. Whether the rate is raised in one step or three, compliance once the new rate is in force is statistically indistinguishable; informing taxpayers of the path in advance reduces full evasion and raises reported income at that same rate.

The policy implication requires care, because it does not follow from comparing revenue across our two arms. Our designs were constructed to carry the same average statutory rate, so which one collects more within the experiment is a consequence of the schedules we chose rather than a behavioral result; Figure 6 shows the stepwise arm collecting more in block 2, when it is at 20% against the abrupt arm's 10%, and less in block 3, when the ordering reverses. The relevant inference runs instead through the compliance result. If arriving at 40% by way of 20% and 30% produces no compliance advantage once 40% is in force, then the intermediate periods purchase nothing: they are revenue forgone in exchange for a behavioral benefit that does not materialize. A government that has determined it needs the higher rate should therefore legislate the higher rate rather than phase it in, and should announce the change in advance, since announcement, unlike pace, measurably reduces the compliance loss that accompanies it.

Figure 6 - Average of tax collection across rounds by reform paths



Note: Tax collection (%) denotes the fraction of generated income collected by the tax authority. In red , average collections under the **stepwise** treatments; in blue , average collections under the **abrupt** treatments. Tax rates by block: **Block 1:** 10% in all treatments. **Block 2:** stepwise 20%, abrupt 10%. **Block 3:** stepwise 30%, abrupt 40%. **Block 4:** 40% in all treatments. Both paths impose an average statutory rate of 25% over the forty periods.

4 Conclusion

We examined how the pace of a tax increase, and advance information about the path it follows, shape individual behavior in a controlled setting. The pace of the reform, whether the rate rises in one step or three, has no detectable influence on effort or on compliance at the terminal rate. Once a reform is approved and implemented, the manner in which it is phased in appears not to be pivotal for how taxpayers respond to it. Advance information about the path, by contrast, mitigates the compliance loss that accompanies the increase: informed participants are 4 to 7 percentage points less likely to declare no income at all, and report roughly 5 percentage points more of their income, at the same 40% rate.[10]

The design also indicates how information operates. In the first block, when every arm faces the same rate and no change has yet occurred, the four arms are indistinguishable and the pooled information effect is essentially zero. Since anticipation of future rates would have to register

[10] The extensive-margin result holds under session-level clustering, individual-level clustering, the wild cluster bootstrap, and randomization inference; the intensive-margin result holds under all but individual-level clustering. See the Online Appendix.

there if participants were reallocating compliance across periods, the effect we observe once rates have risen is not intertemporal optimization. What remains is a motivational channel: participants who were told what was coming meet the same increase with more compliance than participants for whom it arrived as a shock.

The policy implication follows from the compliance result rather than from any revenue comparison within our experiment. Both of our paths carry the same average statutory rate by construction, so which collects more in the laboratory is a property of the schedules we chose, not a behavioral finding. What the experiment establishes is that compliance at the terminal rate does not depend on the route taken to reach it. If arriving at 40% by way of 20% and 30% yields no compliance dividend once 40% is in force, the intermediate periods purchase nothing: they are revenue forgone in exchange for an acceptance benefit that does not materialize. A government that has determined it needs the higher rate should legislate the higher rate rather than phase it in (we observe path-invariance between a three-step increase and a single-step increase taken midway through the experiment, and extrapolate that it would also hold for an increase taken sooner) and should announce the change in advance, since announcement, unlike pace, measurably reduces the compliance loss that accompanies it. This conclusion holds in the environment we construct, in which opposition to a reform can be expressed only through compliance and effort. Where reforms can also be protested, blocked, or reversed, the calculus may differ.

It is worth returning to the advice with which we began. Japan was counselled to raise its consumption tax through stepwise increases, announced in advance, and the two recommendations were presented as complementary ingredients of a single strategy. Our experiment can separate them, and finds that only the second earns its place: announcement mitigates the compliance loss, while gradualism does not. The subsequent Japanese experience speaks to the other margin, the one our design deliberately excludes. The rate never advanced beyond 10%, and reaching even that level required accumulating exemptions and distortions along the way, an outcome consistent with the concern that a protracted path leaves a reform exposed to reversal before completion (Dewatripont and Roland, 1992; Wei, 1997). The two observations point in the same direction from opposite sides. Gradualism's hypothesized benefit does not appear in the domain where we can measure it cleanly, and it carries evident costs in the domain where we cannot.

More broadly, our results locate a boundary for theories of reform sequencing. Those theories concern whether reforms are adopted, and they give good reasons why a slower path might survive political constraints that a faster one would not. What they do not imply, and what our evidence does not support, is that a slower path also purchases taxpayer cooperation once the reform is in force. The two questions are separable, and conflating them, as the four-Ss recommendation does, risks advising governments to pay the costs of gradualism for a benefit that accrues elsewhere, if at all. Our findings also reinforce a point increasingly established in the compliance literature: the deterrence parameters are not the only lever available to a tax authority, and how a change is communicated can matter as much as the change itself

(Hallsworth et al., 2017; Luttmer and Singhal, 2014; Slemrod, 2019).

This paper is just a stepping stone from which to test even more complex reforms and environments. First, our laboratory experiment abstracts from the reform process itself. Providing information about the reforms may increase compliance once they are introduced, but it can also provide fertile ground for more active collective action to begin with. In future research, we aim to introduce another possible action: protesting the reform itself and allowing for the possibility of partial or full reform reversal. This matters for how our policy conclusion should be read: we show that gradualism buys no compliance, not that it buys nothing at all.

Second, in addition to protests, large tax reforms may generate effects on enforcement (Kanbur and Ronconi, 2018). If tax agents, who are taxpayers themselves, believe that the reform is unjust, they may adjust their effort to collect taxes, making the probability of detection endogenous to the tax rate. Future work could randomize the detection and penalties parameters while changing tax rates.

Third, in our laboratory setting the opportunity cost of effort is very low. Participants had no alternative use of their time, so exerting effort weakly dominated regardless of the rate; our design is therefore uninformative about the labor-supply response to taxation, and the null we report on effort should not be read as evidence that none exists.

Fourth, while we vary the pace of the change, limited power prevented us from randomizing its depth. Future work could vary the size of the individual steps rather than only their number, which would additionally reduce the scope for participants to learn the pattern of increases and would raise uncertainty about what comes next, closer to the conditions under which the monetary policy literature finds that responses to policy surprises depend on the prevailing level of uncertainty (De Pooter et al., 2018).

Finally, our participants are university students in a single laboratory, deciding over stakes of roughly twelve dollars, and the taxes they pay are directed to an organization of their choosing rather than to a government. Laboratory compliance experiments reproduce the qualitative comparative statics found in field data, and students respond to the policy parameters we manipulate much as actual taxpayers do (Alm et al., 2015; Choo et al., 2016; Mascagni, 2018). We nonetheless read our estimates as evidence on the direction and relative magnitude of the two treatments rather than as calibrated predictions of field responses, and the extrapolation to real reforms, where announcements are made by governments with reputations, and where taxpayers can respond politically as well as fiscally, should be made with corresponding caution.

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Gradual or Abrupt?
Reform Pace, Advance Information, and Tax Compliance

Online Appendix

A.1 Theoretical motivation

A.1.1 A static benchmark

We modify the standard tax evasion model of Allingham and Sandmo (1972) and Srinivasan (1973) to include an effort component for income generation and an intrinsic motivation parameter for paying taxes (Dwenger et al., 2016). An individual decides how much income to generate, z (think of z as $w \cdot l$, where w is the market wage rate and l is labor), and how much income $y \leq z$ to declare, on which she pays a tax rate τ. Utility has three components. It increases in after-tax income transformed into consumption ($U'(c) > 0$, $U''(c) < 0$); it decreases in the effort required to produce income ($U'(z) < 0$, $U''(z) < 0$) for given consumption; and the individual holds an intrinsic motivation to report income and pay taxes ($S'(y) > 0$, $S''(y) < 0$, $U'(S(y)) > 0$, $U''(S(y)) < 0$).

The individual is audited with probability p, and if audited pays a fine $f > 1$ on the evaded tax. Her problem is to maximize:

$$E\left[U\left(c, z, S\left(y\right)\right)\right] \equiv \underbrace{(1-p)U\left(z - y\tau, z, S\left(y\right)\right)}_{\substack{\text{Expected return} \\ \text{of not being audited}}} + \underbrace{pU\left(z - y\tau - f\tau(z-y), z, S\left(y\right)\right)}_{\substack{\text{Expected return} \\ \text{of being audited}}}$$

The comparative statics are standard: higher penalties, a higher detection probability, and stronger intrinsic motivation all increase declared income. An increase in the tax rate has an ambiguous effect on declaration—it raises the gain from evasion but also the cost if audited—and a negative effect on income generation through reduced post-tax marginal returns to effort. Note that nothing in this benchmark depends on the history of tax rates: declaration is a function of the rate currently in force, not of the path by which it was reached. Establishing what would have to be added to the model for the path to matter, and testing whether it does, is the purpose of what follows.

A.1.2 Adding a path of tax rates

Introducing information about the future path of rates requires a finite-horizon version. Let z_t denote income generated in period t, $y_t \leq z_t$ income declared, τ_t the tax rate, p_t the audit probability, $f > 1$ the penalty multiplier, and $\delta \in (0, 1]$ the discount factor. Consumption depends on the audit outcome:

$$c_t = z_t - \tau_t y_t \quad \text{(not audited)}$$
$$c_t^{\text{audit}} = z_t - \tau_t y_t - f\tau_t(z_t - y_t) \quad \text{(audited)}$$

so that period-t expected utility is $EU_t = (1 - p_t)U(c_t, z_t, S(y_t, \tau_t)) + p_t U(c_t^{\text{audit}}, z_t, S(y_t, \tau_t))$, and the individual chooses $\{z_t, y_t\}_{t=1}^T$ to maximize

$$\max_{\{z_t, y_t\}_{t=1}^T} \sum_{t=1}^{T} \delta^{t-1} \left[(1 - p_t)U(z_t - \tau_t y_t, z_t, S(y_t, \tau_t)) + p_t U(z_t - \tau_t y_t - f\tau_t(z_t - y_t), z_t, S(y_t, \tau_t))\right]$$

Informed and uninformed individuals differ in their information set. The informed optimize with full knowledge of $\{\tau_s\}_{s=1}^T$ from $t = 1$. The uninformed form expectations $\mathbb{E}[\tau_s]$; since they have no reason to anticipate a change, we take $\mathbb{E}[\tau_s] = \tau_1$ for all s, so that in each period they re-optimize myopically under the belief that the current rate will persist.

Within any period, a higher rate has two opposing effects on declaration. It raises the gain from evasion, since each unit of undeclared income saves τ, which pushes y down. It also raises the fine conditional on detection, which pushes y up. Which dominates depends on risk aversion: at low levels of risk aversion, evasion rises with the rate; at high levels, it falls. Laboratory evidence with modest audit penalties generally finds the former.

A.1.3 Does advance information change first-period behavior?

Whether informed and uninformed individuals should behave differently before any change occurs depends on what links periods to one another. Three candidate links are absent here by design. There is no intertemporal substitution of the tax base: income is generated each block afresh and cannot be shifted across periods. There is no accumulation: our payment protocol pays only two randomly drawn rounds per block, so wealth realized in one round does not enter the budget constraint of any other. And there is no learning about enforcement, since p_t and f are constant and announced.

One link nevertheless remains. The individual is paid on the sum of the drawn rounds, so if she evaluates the experiment as a whole rather than round by round, each round's audit gamble

is background risk for every other. This generates a signed prediction. Informed participants know at $t = 1$ that future rates will be higher, and hence that both their anticipated total earnings are lower and their future audit exposure is larger, since the penalty $f\tau_t(z_t - y_t)$ is proportional to the rate. Under decreasing absolute risk aversion, lower anticipated wealth raises the individual's aversion to the current gamble; under risk vulnerability, so does greater background risk (Gollier and Pratt, 1996). Both considerations push in the same direction:

$$y_1^{\text{Info}} \geq y_1^{\text{No Info}}$$

with equality if the two are not linked at all. This inequality is what makes the first block a discriminating test rather than a balance check. It is the only prediction in the model under which advance information affects behavior before any tax change has occurred.

Two considerations suggest the effect should be small. First, at total stakes of roughly $12 USD, the wealth differential between informed and uninformed participants at $t = 1$ is a few dollars in expectation, and the induced change in absolute risk aversion is negligible for any plausible utility function. Second, the protocol presents each round as a self-contained decision followed by its own audit lottery and its own result screen, which encourages narrow bracketing (Read et al., 1999); under narrow bracketing the link disappears entirely, and the inequality holds with equality. We therefore expect the first-block difference to be close to zero, and we test it directly rather than assuming it.

A.1.4 History dependence in tax morale

The channels above all concern optimization. A second possibility is that the path of rates affects the intrinsic motivation to comply. Suppose that when the rate rises from τ to $\tau + \Delta\tau$, the individual does not immediately perceive the new rate as her reference point, but instead perceives $\tau + \theta\Delta\tau$ with $\theta \in (0, 1]$, so that intrinsic motivation becomes $S(y, \tau + \theta\Delta\tau)$. The parameter θ governs how quickly the reference rate adapts to a change.

If $\theta = 1$, perception tracks the statutory rate exactly, intrinsic motivation is unaffected by the history of the reform, and declared income in equilibrium at any given rate is the same regardless of the path taken to reach it. If $\theta < 1$, a recent increase is perceived as smaller than it is, and the two designs are no longer equivalent at the terminal rate: under the abrupt design the last change occurred one block earlier and was a thirty-point jump, whereas under the stepwise design it occurred in the current block and was a ten-point step. The designs therefore differ in both the size and the recency of the final adjustment, and any difference in block-4 compliance between them identifies $\theta \neq 1$.

This is the parameter that theories of gradualism implicitly assume to be less than one. The argument that a slower reform path eases acceptance requires that taxpayers adapt to intermediate rates in a way that carries over to the terminal rate; in the language of this model, it

requires that arriving at 40% through 20% and 30% leaves the individual perceiving something lower than arriving there directly. The same logic underlies reference-dependent accounts of tax behavior (Engström et al., 2015) and, at a different level of aggregation, models in which gradual reform builds the constituencies that sustain it (Dewatripont and Roland, 1992; Wei, 1997). Testing whether $\theta = 1$ is therefore a test of the behavioral premise on which advice to phase in reforms rests.

Advance information may operate on the same object. If disclosure of the reform path raises trust in the authority while unannounced increases erode it, then $S(\cdot)$ is higher for informed individuals once a change has been delivered, though not before it (Feld and Frey, 2007; Keefer et al., 2021; Luttmer and Singhal, 2014).

A.1.5 Hypotheses

[H1] Baseline equivalence. In block 1, before any change has occurred, uninformed participants should behave identically regardless of the reform path they will subsequently face:

$$y_1^{\text{Abrupt, No Info}} \simeq y_1^{\text{Stepwise, No Info}}$$

[H2] Rate effect. For the uninformed, who re-optimize each period believing the current rate will persist, compliance should fall as the rate rises, unless risk aversion is high:

$$y_1^{\text{No Info}} \geq y_2^{\text{No Info}} \geq y_3^{\text{No Info}} \geq y_4^{\text{No Info}}$$

[H3] Anticipation. In block 1, informed participants face the same rate as uninformed participants but anticipate lower future earnings and greater future audit exposure. If they integrate outcomes across rounds, decreasing absolute risk aversion and risk vulnerability imply higher compliance; if they bracket narrowly, no difference:

$$y_1^{\text{Abrupt, Info}} \geq y_1^{\text{Abrupt, No Info}}$$

$$y_1^{\text{Stepwise, Info}} \geq y_1^{\text{Stepwise, No Info}}$$

This is the only prediction under which information affects behavior before a change has occurred, and it therefore separates the intertemporal channel from the motivational one.

[H4] Pace. At the terminal rate, compliance is independent of the path taken to reach it if and only if intrinsic motivation is history-independent, that is, if $\theta = 1$:

$$y_4^{\text{Abrupt}} \simeq y_4^{\text{Stepwise}} \quad \Longleftrightarrow \quad \theta = 1$$

A difference between the designs identifies $\theta \neq 1$ and would support the behavioral premise underlying advice to phase in reforms.

[H5] Information and tax morale. If disclosure raises trust and unannounced increases erode it, informed participants should comply more once the terminal rate is in force, under either design:

$$y_4^{\text{Abrupt, Info}} \geq y_4^{\text{Abrupt, No Info}}$$
$$y_4^{\text{Stepwise, Info}} \geq y_4^{\text{Stepwise, No Info}}$$

[H6] Effort. The marginal cost of effort is unchanged by the tax rate, while the marginal benefit of labor falls proportionally as the rate rises, so effort should decline with the rate.[11]

[11] The model and our implementation differ in one respect here. The model assumes effort carries an opportunity cost and hence a disutility. In the laboratory, participants had no alternative use of their time, so exerting effort weakly dominates at any rate, and we do not expect H6 to hold in our setting. We report the effort results for completeness rather than as a test.

A.2 Additional Tables

A.2.1 Balance Table

Table A.1 Balance between treatment groups

Dep Var: Covariate of interest	Sociodemographic variables							
	Age	Female	Expenses	Risk measure	Stratum	Political preference	Graduated	Grad. Level
Abrupt with information (T_1)	0.675	0.078	1.413	-0.186	-0.042	0.254	0.014	-0.013
	(0.450)	(0.070)	(11.471)	(0.281)	(0.137)	(0.174)	(0.048)	(0.041)
Stepwise with no information (T_2)	-0.246	-0.004	-3.035	-0.151	-0.020	-0.084	0.016	0.026
	(0.390)	(0.070)	(8.906)	(0.279)	(0.142)	(0.185)	(0.048)	(0.044)
Stepwise with information (T_3)	0.255	0.115	-9.345	-0.021	-0.037	0.016	0.011	-0.029
	(0.528)	(0.070)	(7.785)	(0.273)	(0.135)	(0.176)	(0.048)	(0.039)
Constant	21.535***	0.465***	44.323***	6.891***	3.347***	3.574***	0.129***	0.099***
	(0.265)	(0.050)	(7.203)	(0.188)	(0.093)	(0.125)	(0.033)	(0.030)
p-value $H_0 : T_1 = T_2$	0.047	0.241	0.668	0.903	0.880	0.064	0.978	0.357
p-value $H_0 : T_1 = T_3$	0.472	0.594	0.253	0.566	0.970	0.169	0.953	0.676
p-value $H_0 : T_2 = T_3$	0.352	0.090	0.295	0.650	0.907	0.589	0.931	0.185
Observations	410	410	410	410	410	410	410	410
Adjusted R-squared	0.01	0.01	0.00	0.00	0.00	0.01	0.00	0.00

Notes: This table shows balance between the treatment groups and the control group (abrupt with no information). The treatments differ by the changes in the information structure and tax rate. *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. At the bottom, we present the p-values for the difference in means between treatment groups.

$^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$

A.2.2 Real Effort Tasks

Table A.2 - Effect of tax rate over effort on the RET

Dep Var: **N° of encrypted words**	*Real Effort Task*				
	General Sample	Abrupt		Stepwise	
		No information	Information	No information	Information
First task (ϵ_1)	1.449***	1.426**	1.229**	1.760***	1.380***
	(0.128)	(0.292)	(0.325)	(0.275)	(0.187)
Second task (ϵ_2)	2.301***	2.248***	2.105**	2.837***	2.042***
	(0.162)	(0.294)	(0.363)	(0.351)	(0.154)
Third task (ϵ_3)	2.557***	2.208***	2.324*	3.279***	2.457***
	(0.241)	(0.340)	(0.797)	(0.328)	(0.206)
Fourth task (ϵ_4)	2.611***	2.406***	2.448***	3.077***	2.561***
	(0.136)	(0.292)	(0.256)	(0.308)	(0.170)
Mean Dep. Variable	21.97	22.27	22.58	21.17	21.84
Controls	✓	✓	✓	✓	✓
Clusters	16	4	4	4	4
Observations	2041	505	525	520	491
Adjusted R-squared	0.05	0.07	0.10	0.12	0.08
p-value $H_0 : \epsilon_1 = \epsilon_2$	0.000	0.002	0.040	0.006	0.094
p-value $H_0 : \epsilon_1 = \epsilon_3$	0.000	0.042	0.153	0.003	0.058
p-value $H_0 : \epsilon_1 = \epsilon_4$	0.000	0.003	0.032	0.022	0.010
p-value $H_0 : \epsilon_2 = \epsilon_3$	0.095	0.857	0.701	0.007	0.032
p-value $H_0 : \epsilon_2 = \epsilon_4$	0.018	0.190	0.162	0.493	0.136
p-value $H_0 : \epsilon_3 = \epsilon_4$	0.756	0.205	0.851	0.473	0.741
WCB p (ϵ_1)	0.000	0.043	0.057	0.039	0.036
WCB p (ϵ_2)	0.000	0.034	0.029	0.032	0.024
WCB p (ϵ_3)	0.000	0.032	0.161	0.021	0.027
WCB p (ϵ_4)	0.000	0.031	0.024	0.025	0.015

Notes: The baseline category is the *Baseline task*. The first through fourth tasks correspond to RET 1 through RET 4, denoted by (ϵ_1) to (ϵ_4), respectively. In the **Abrupt** treatment, tasks ϵ_1 and ϵ_2 were conducted under a tax rate of 10%, while ϵ_3 and ϵ_4 were conducted under 40%. In the **Stepwise** treatment, ϵ_2, ϵ_3, and ϵ_4 were conducted under tax rates of 20%, 30%, and 40%, respectively. All specifications control for gender, age, socioeconomic stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization; 16 sessions, or 4 per treatment arm) are presented in parentheses. The row *WCB p-value* reports wild cluster bootstrap p-values (Webb weights, 9,999 replications) for each task coefficient, which account for the small number of clusters. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$

A.2.3 Tax Compliance

Note: Tax collection (%) denotes the fraction of endowment collected by the national authority. In squares (■), the average compliance rate for **abrupt** treatments. In circles (●), the average compliance rate for **stepwise** treatments. Dashed, **informed** treatments. Solid, **uninformed** treatments. **Left panel:** denotes choices when tax rate is 10% and **Right panel:** denotes choices when tax rate is 40%.

Table A.3 - Treatment effect over effort by RET.

Dep Var: **N° of encrypted words**	*Real Effort Task*				
	(Baseline)	(Block 1)	(Block 2)	(Block 3)	(Block 4)
Abrupt with information (T_1)	0.380	-0.044	-0.026	0.210	0.115
	(0.380)	(0.416)	(0.408)	(0.805)	(0.342)
Stepwise with no information (T_2)	-1.110**	-0.014	0.251	0.748*	0.358
	(0.381)	(0.301)	(0.305)	(0.351)	(0.387)
Stepwise with information (T_3)	-0.342	-0.168	-0.344	0.105	0.008
	(0.453)	(0.283)	(0.302)	(0.374)	(0.314)
Mean Dep. Variable	22.27	23.69	24.51	24.48	24.67
Controls	✓	✓	✓	✓	✓
Baseline performance control		✓	✓	✓	✓
Observations	410	410	409	406	406
Adjusted R-squared	0.03	0.54	0.58	0.49	0.49
p-value H_0: $T_1 = T_2$	0.002	0.951	0.551	0.515	0.623
p-value H_0: $T_1 = T_3$	0.150	0.780	0.463	0.898	0.780
p-value H_0: $T_2 = T_3$	0.130	0.653	0.122	0.117	0.398
WCB p (Abrupt Info)	0.376	0.928	0.954	0.787	0.763
WCB p (Step No Info)	0.022	0.962	0.420	0.036	0.373
WCB p (Step Info)	0.488	0.577	0.385	0.812	0.978

Note: All specifications control for gender, age, socioeconomic stratum, risk aversion, weekly expenses, and political preferences. The *Baseline performance control* denotes the inclusion of the baseline performance as control for the estimation. The *Mean Dep. Variable* expresses the unconditional mean of the dependent variable for the control group: *Abrupt with no information*. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses. The *WCB p-value* rows report wild cluster bootstrap p-values (Webb weights, 9,999 replications) for each treatment coefficient, which account for the small number of clusters. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

Table A.4 Effect of the information treatment over **compliance** rate – individual-level clustering

Dep Var: **Compliance rate**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information treatment	0.005	-0.002	-0.002	0.001	0.054*	0.049*	0.049*	0.061**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Mean dep. var	0.63	0.63	0.63	0.63	0.50	0.50	0.50	0.50
Effect size (%)	0.8	-0.3	-0.3	0.1	10.9	9.7	9.7	12.1
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4090	3681	3681	3681	4040	4040	4040	4040
Clusters	409	409	409	409	404	404	404	404
Adjusted R-squared	0.00	0.03	0.03		0.00	0.05	0.05	

Notes: Same specifications as Table 7, but with standard errors clustered at the individual level (the number of clusters equals the number of participants in the estimation sample) instead of the session level. Compare to Table 7 in the main text.

Table A.5 Effect of the information treatment over **full evasion** – individual-level clustering

Dep Var: **Full evasion**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information treatment	-0.028	-0.018	-0.018	-0.022	-0.074**	-0.054*	-0.054*	-0.061**
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
Mean dep. var	0.09	0.09	0.09	0.09	0.19	0.19	0.19	0.19
Effect size (%)	-30.3	-19.5	-19.5	-22.8	-37.9	-27.9	-27.9	-31.3
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	4100	4100	4100	3690	4060	4060	4060	4060
Clusters	410	410	410	410	406	406	406	406
Adjusted R-squared	0.00	0.03	0.03		0.01	0.07	0.07	

Notes: Same specifications as Table 6, but with standard errors clustered at the individual level (the number of clusters equals the number of participants in the estimation sample) instead of the session level. Compare to Table 6 in the main text.

Figure A.1 - Average of compliance rate across rounds by treatment

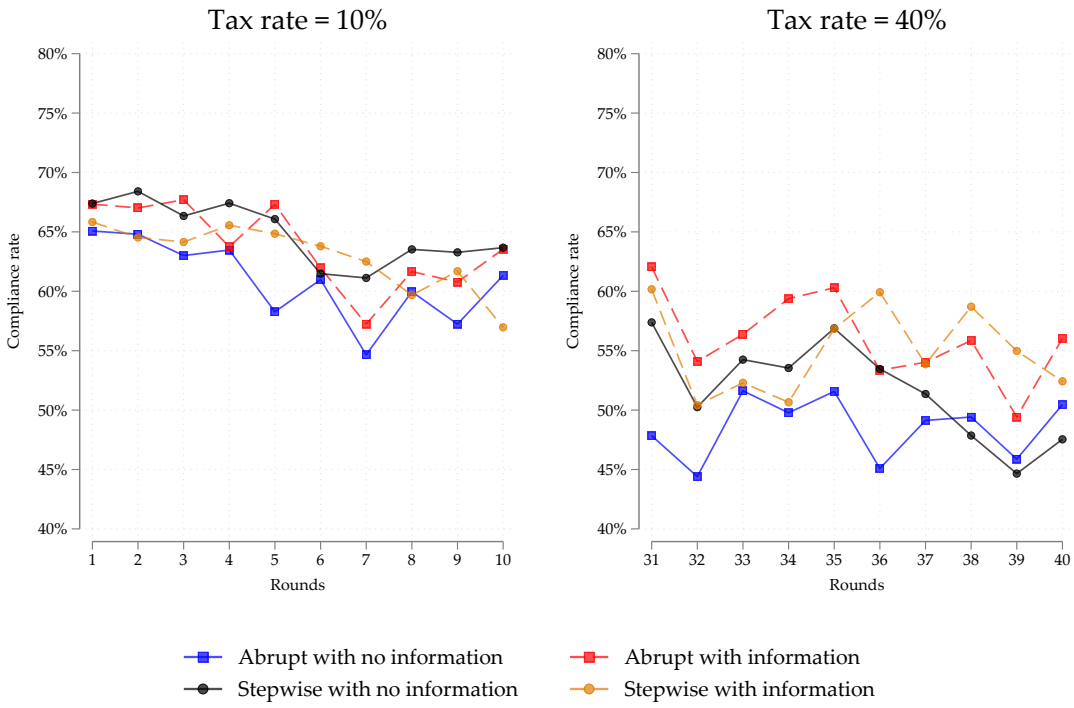


Table A.6 Randomization testing effect of information over **Compliance**

Dep Var: **Compliance rate**	Block	
	Tax rate = 10% result$_reg_1$	Tax rate = 40% result$_reg_4$
Information treatment	0.005	0.054**
	(0.026)	(0.020)
Constant	0.629***	0.501***
	(0.016)	(0.014)
Randomization inference **p-value**	0.85	0.02
N° of samples **higher** than observed	847	21
N° of samples (reps)	1000	1000

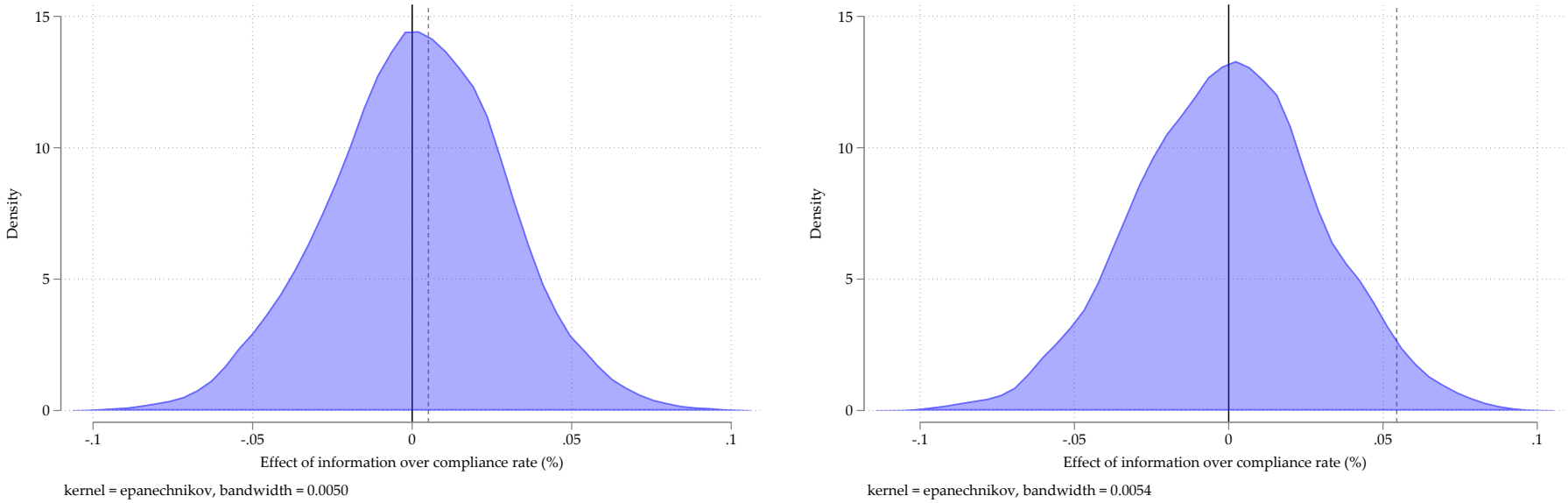
Notes: We employed randomization testing as a robustness check analysis. Since treatment was assigned at the session level, we resample entire sessions (not individuals) to their treatment status, matching the actual randomization design, and compute 1,000 possible *t distributions*; we then compare the observed effect with the simulated effect. We employed the STATA command *ritest*, clustering both the estimation and the resampling at the session level, to compute the Monte Carlo simulations. We do not include controls for our estimation.

Table A.7 Starting and Ending effect - Robustness Check (**Rounds 2 - 38**)

Dep Var: **Compliance rate**	Block							
	Tax rate = 10%				Tax rate = 40%			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Information treatment	0.007	-0.001	-0.001	0.008	0.053**	0.028***	0.028***	0.052**
	(0.03)	(0.01)	(0.01)	(0.03)	(0.02)	(0.01)	(0.01)	(0.02)
Mean dep. var	0.62	0.62	0.62	0.62	0.51	0.51	0.51	0.51
Effect size (%)	1.1	-0.2	-0.2	1.3	10.4	5.5	5.5	10.2
Controls		✓	✓			✓	✓	
Round FE			✓	✓			✓	✓
LASSO Controls				✓				✓
Observations	3272	3272	3272	3272	3232	3231	3231	3232
Clusters	16	16	16	16	16	16	16	16
Adjusted R-squared	0.00	0.30	0.30		0.00	0.27	0.28	

Notes: *Baseline (abrupt - no information):* The tax rate is 10% for the first two blocks (periods 1-20) and 40% for the remaining two blocks (periods 21-40). *Treatment 1 (abrupt - information):* The change in the tax rate is the same as the Baseline, but before the first block, the subjects are informed about the tax rate for each period. *Treatment 2 (stepwise - no information):* The tax rate is 10% for the first block (periods 1-10), 20% for the second block (periods 11-20), 30% for the third block (periods 21-30), and 40% for the remaining block (periods 31-40). *Treatment 3 (stepwise - information):* The change in the tax rate is the same as Treatment 2, but before the first block, the subjects are informed about the tax rate for each period. We include controls for female, age, stratum, risk aversion, weekly expenses, and political preferences. Standard errors clustered at the session level (the unit of randomization; 16 sessions) are presented in parentheses.

Figure A.2 Randomization inference analysis

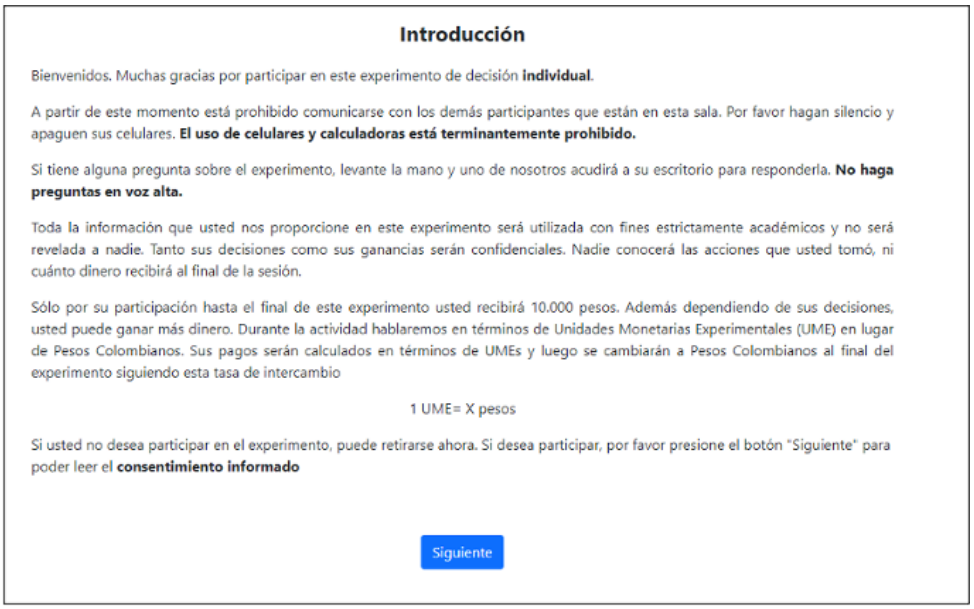


(a) Treatment effect with tax rate of **10%**

(b) Treatment effect with tax rate of **40%**

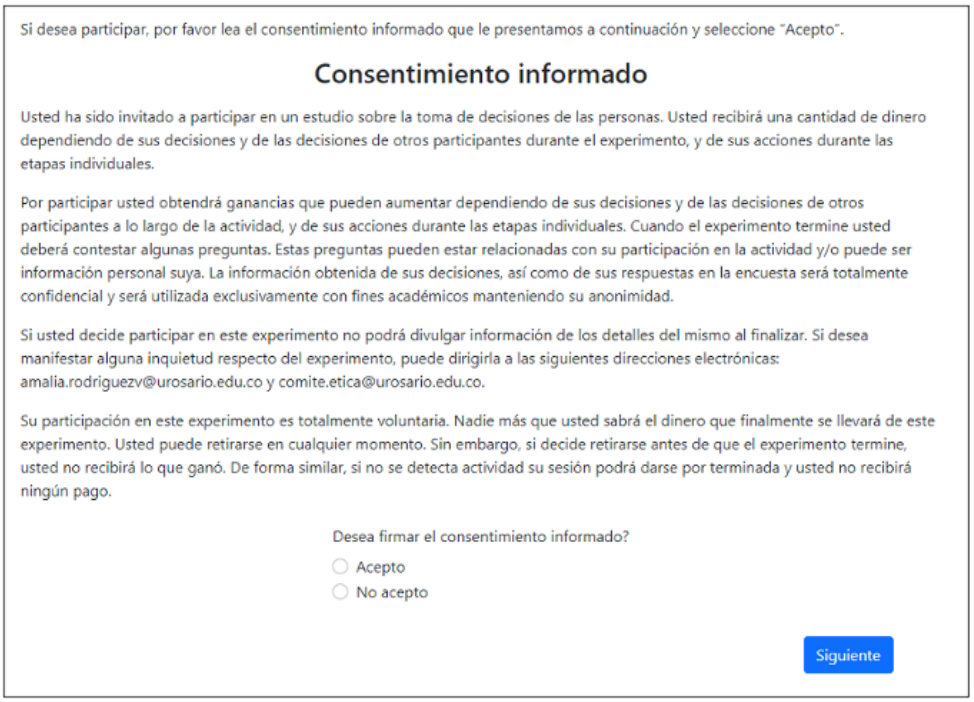
Note: Solid vertical line marks the true estimated treatment effect from the observed data; dashed line marks the two-tailed 5% critical value. The density plots the distribution of placebo treatment effects obtained by randomly reassigning treatment status across units 1,000 times (holding sample size and design fixed) and re-estimating the effect under each reshuffled assignment. The randomization-inference p-value is the share of these placebo estimates at least as large in absolute value as the true effect; if this exceeds conventional significance thresholds, the observed effect cannot be distinguished from what random reassignment alone would produce. Panel (a) is estimated for 10% tax rate, Panel (b) is estimated for 40% tax rate.

Figure A.3 - Screen – Introduction



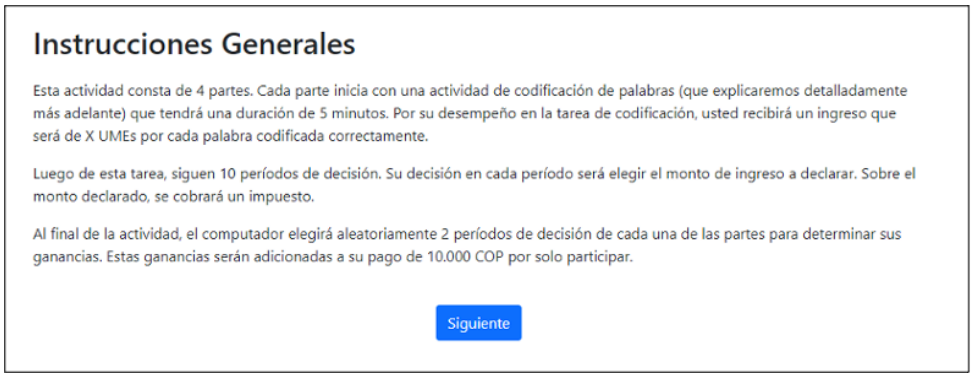
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.4 - Screen – Informed Consent



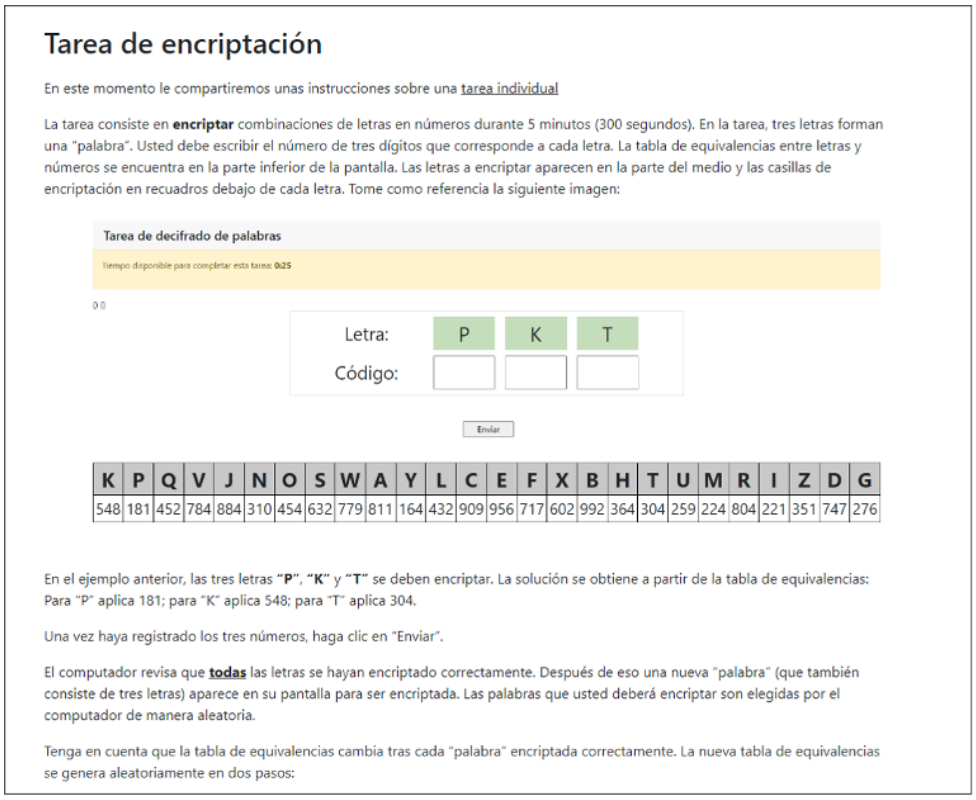
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.5 - Screen – General Instructions



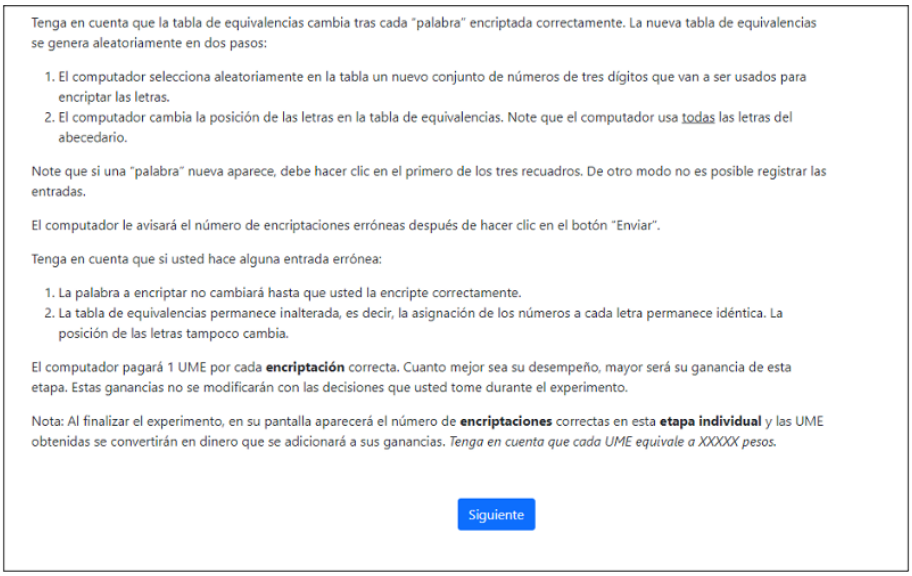
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.6 - Screen – Encryption Task Instructions I



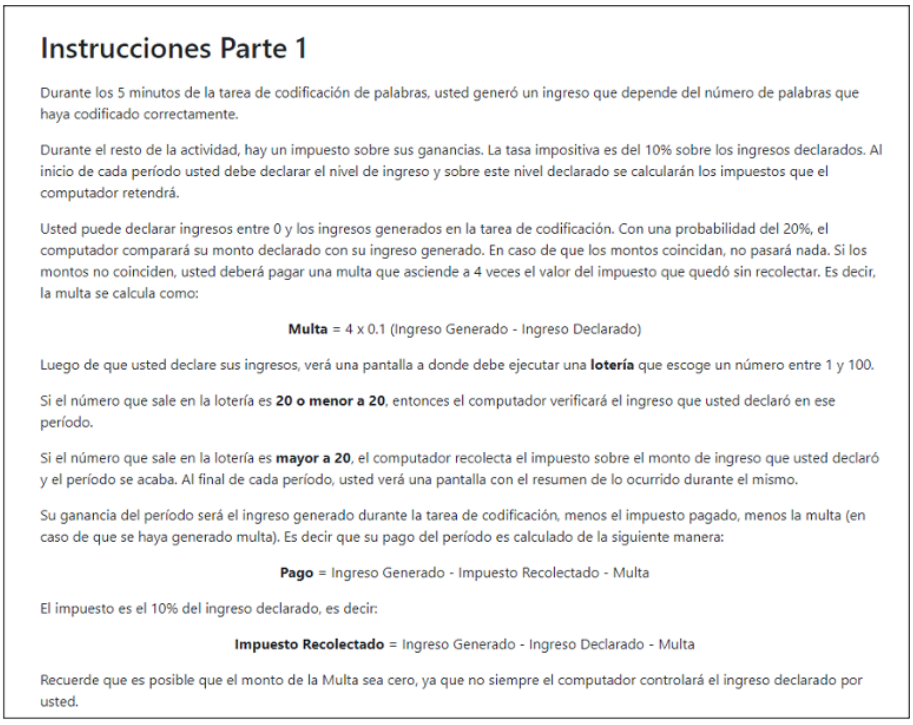
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.7 - Screen – Encryption Task Instructions II



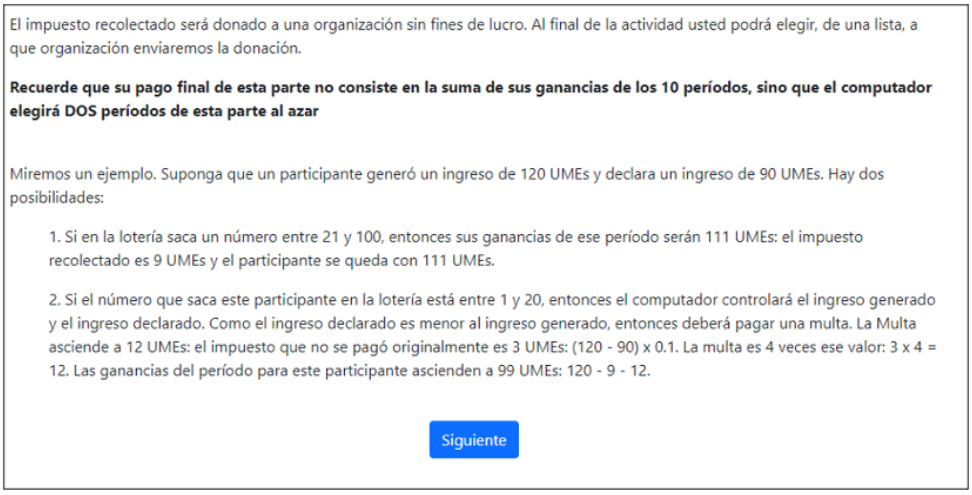
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.8 - Screen – Tax Declaration Instructions I



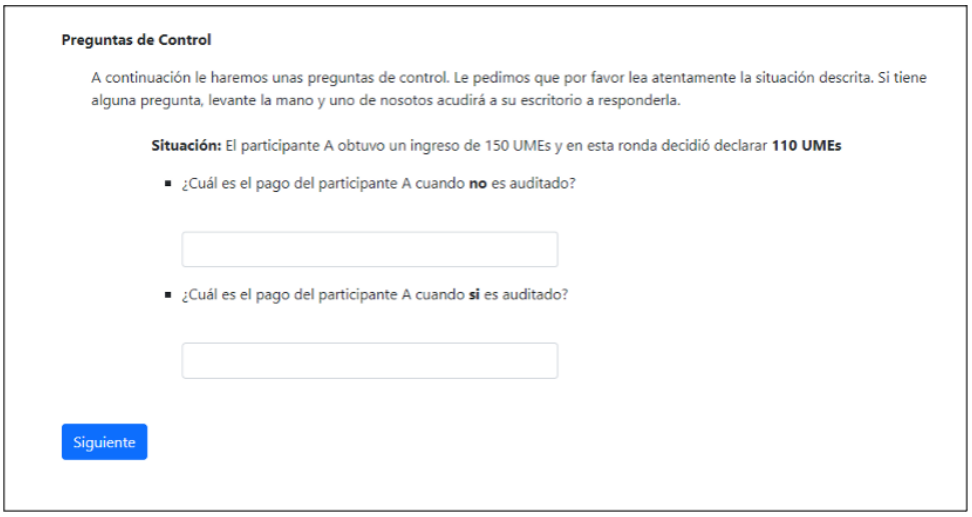
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.9 - Screen – Tax Declaration Instructions II



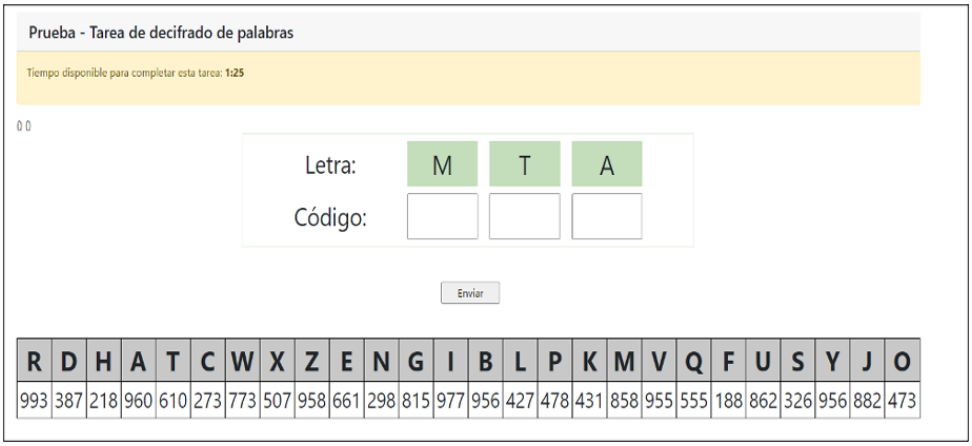
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.10 - Screen – Control Question



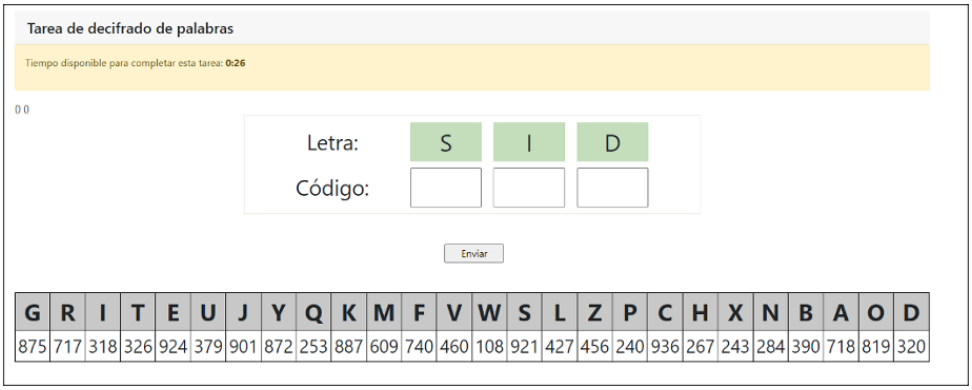
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.11 - Screen – Encryption Task



Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.12 - Screen – Encryption Result



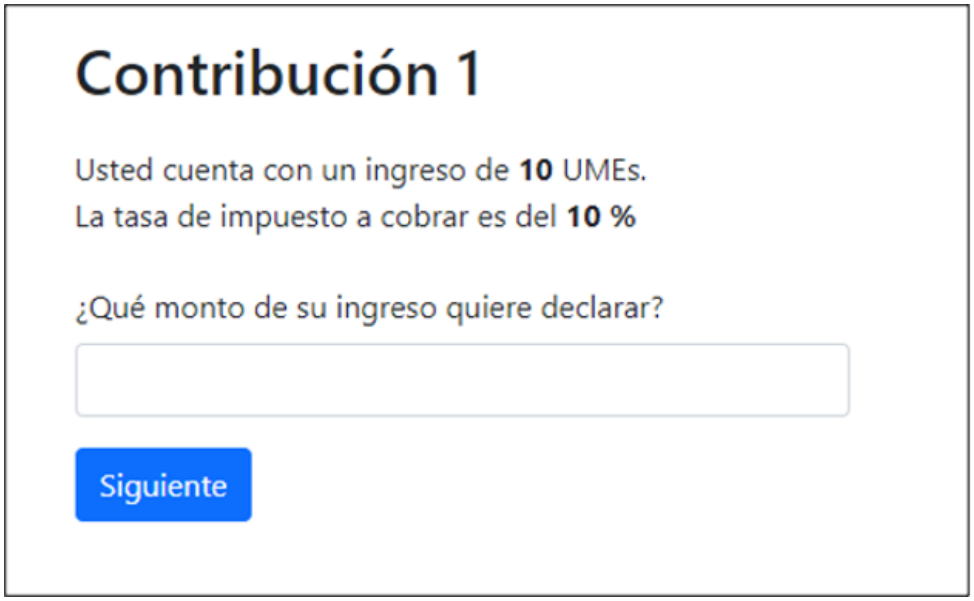
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.13 - Screen – Results of the Encryption Task



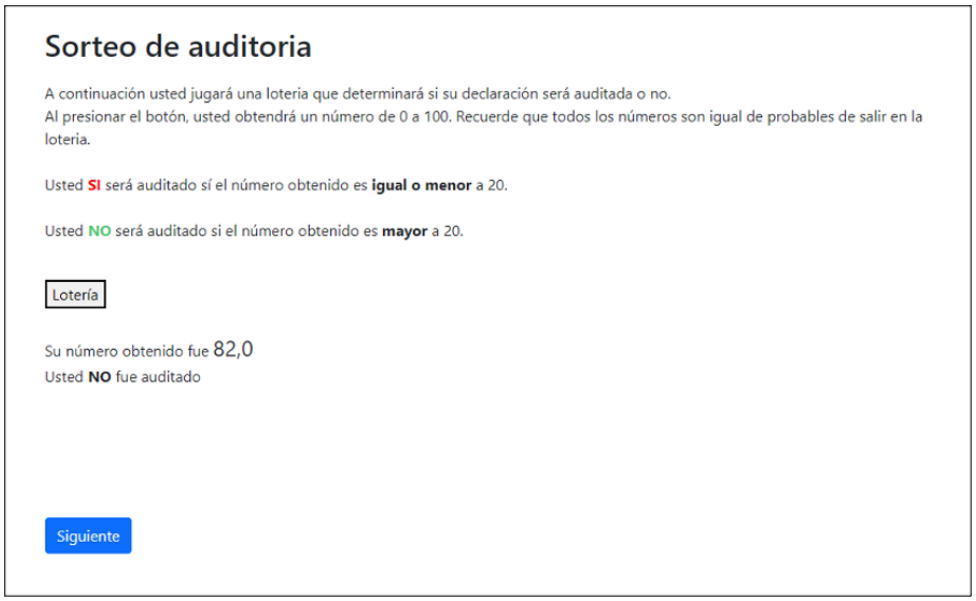
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.14 - Screen – Declaration



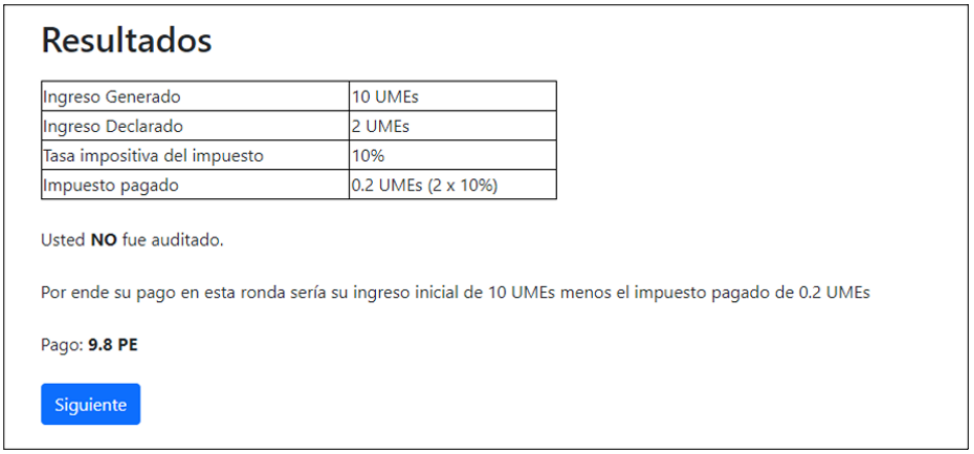
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.15 - Screen – Audit Lottery



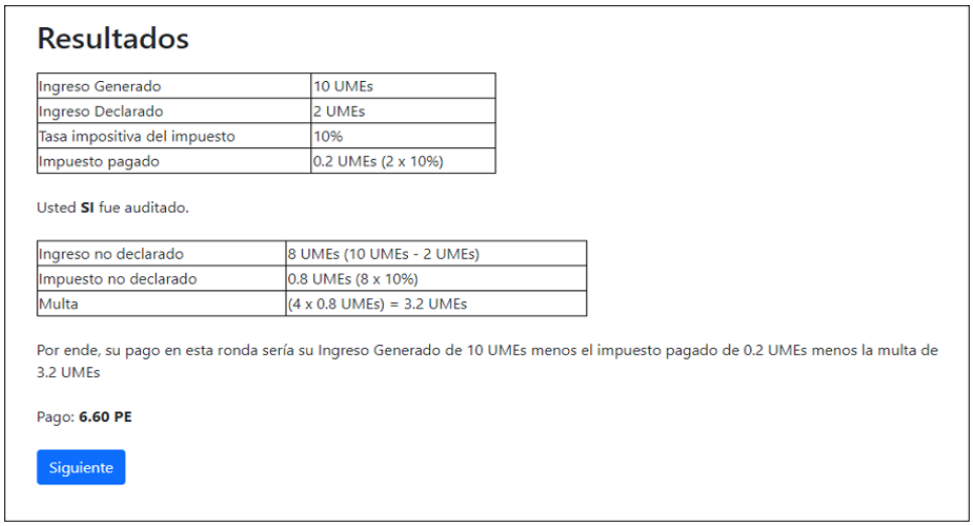
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.16 - Screen – Results When Not Audited



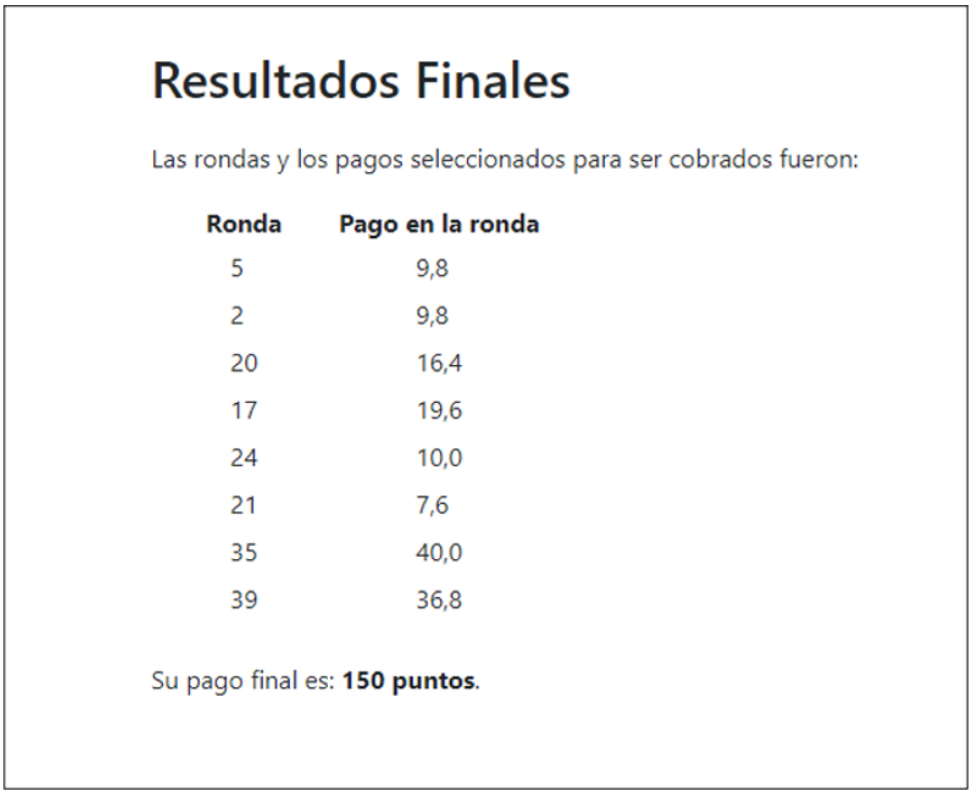
Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.17 - Screen – Results When Audited



Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.

Figure A.18 - Screen – Final Results



Note: The experiment was conducted using oTree (Chen et al., 2016) involving 410 college students from various universities in Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL) between November 4, 2022, and March 2, 2023. The instructions are in Spanish.